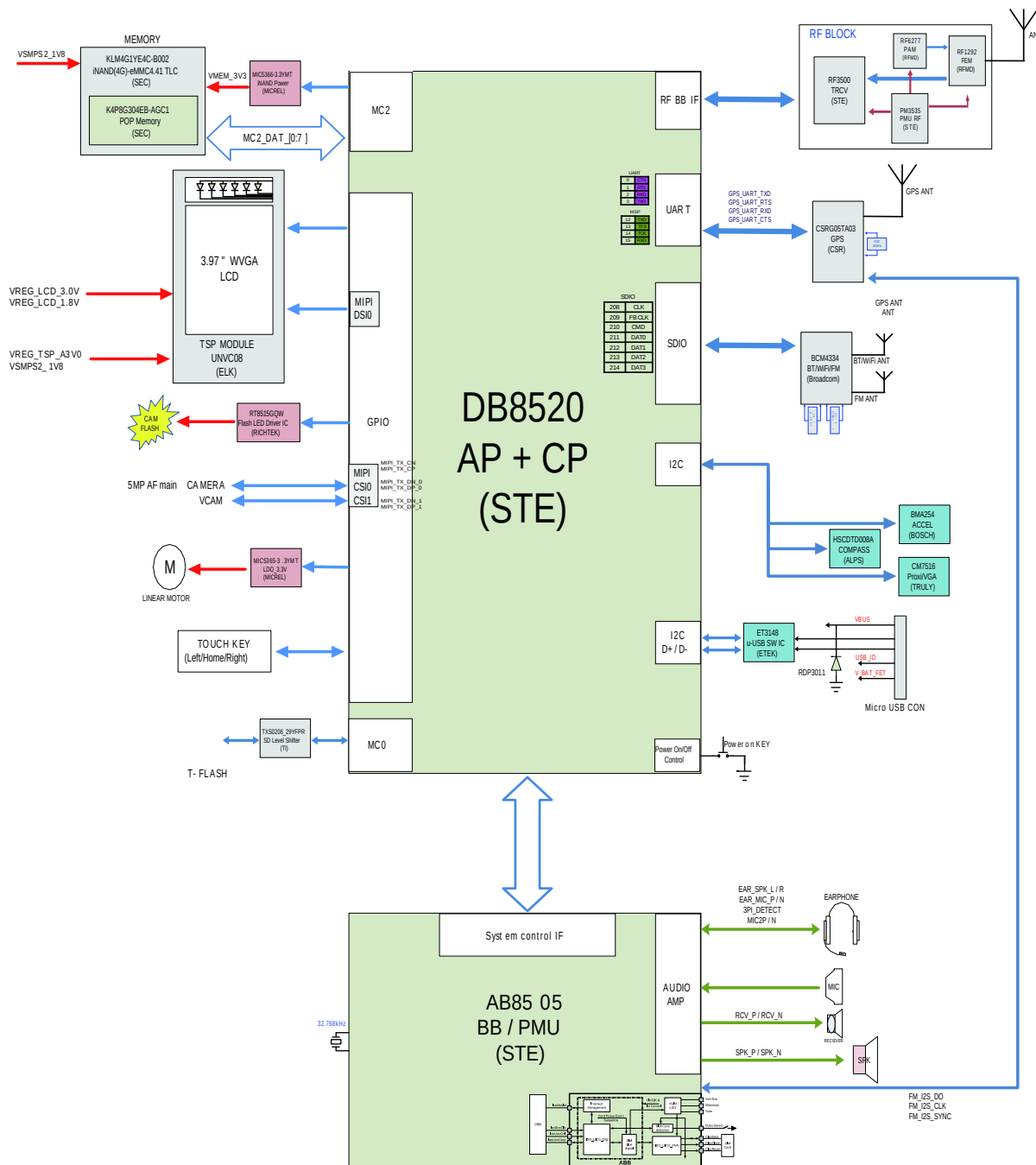


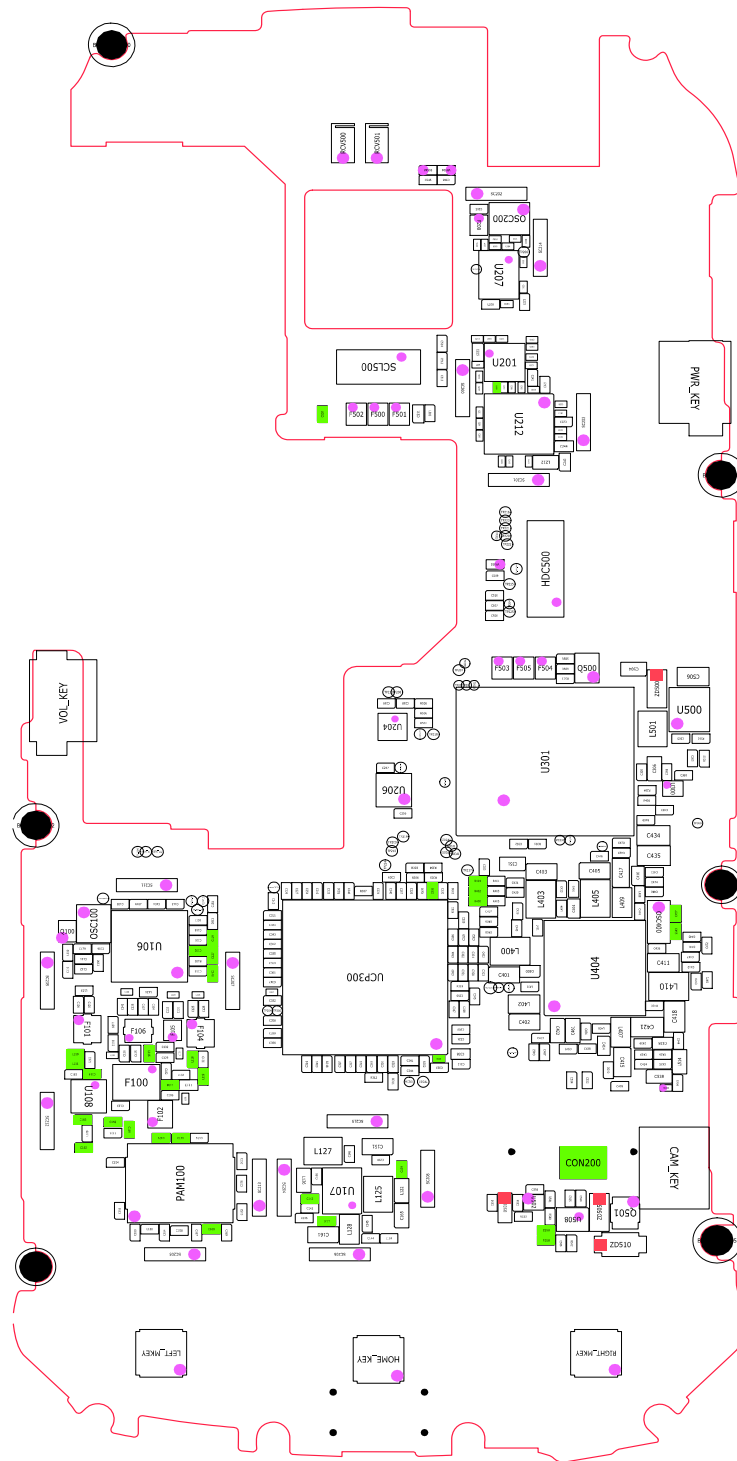
8. Level 3 Repair

8-1. Block Diagram

Updated date : 2013.01.18

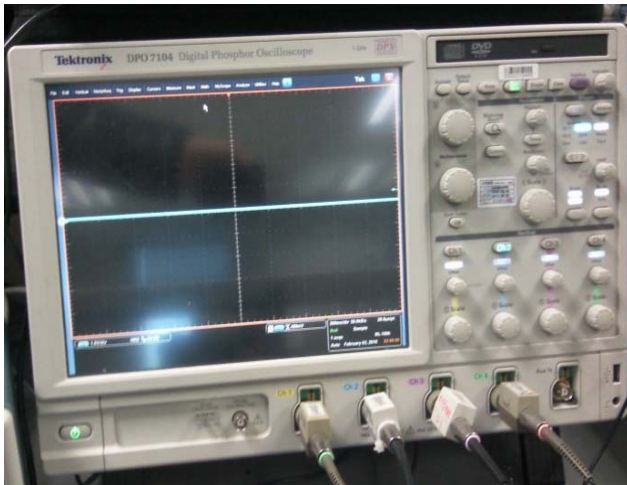
GT-S7710 BLOCK DIAGRAM





8-3. Flow Chart of Troubleshooting

Equipments



↑ Oscilloscope



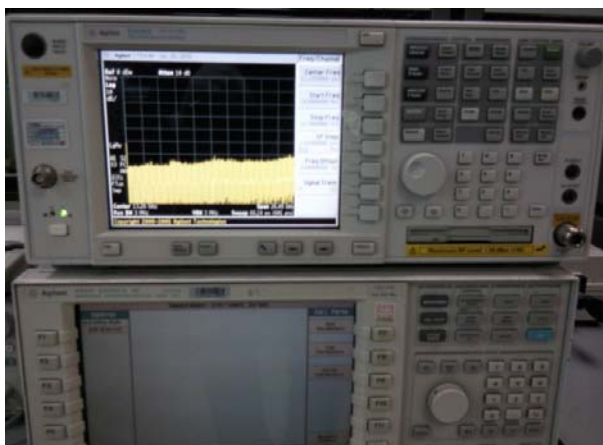
↗ Digital Multi-meter



↗ Power Supply



↗ + driver, ESD Safe Tweezer

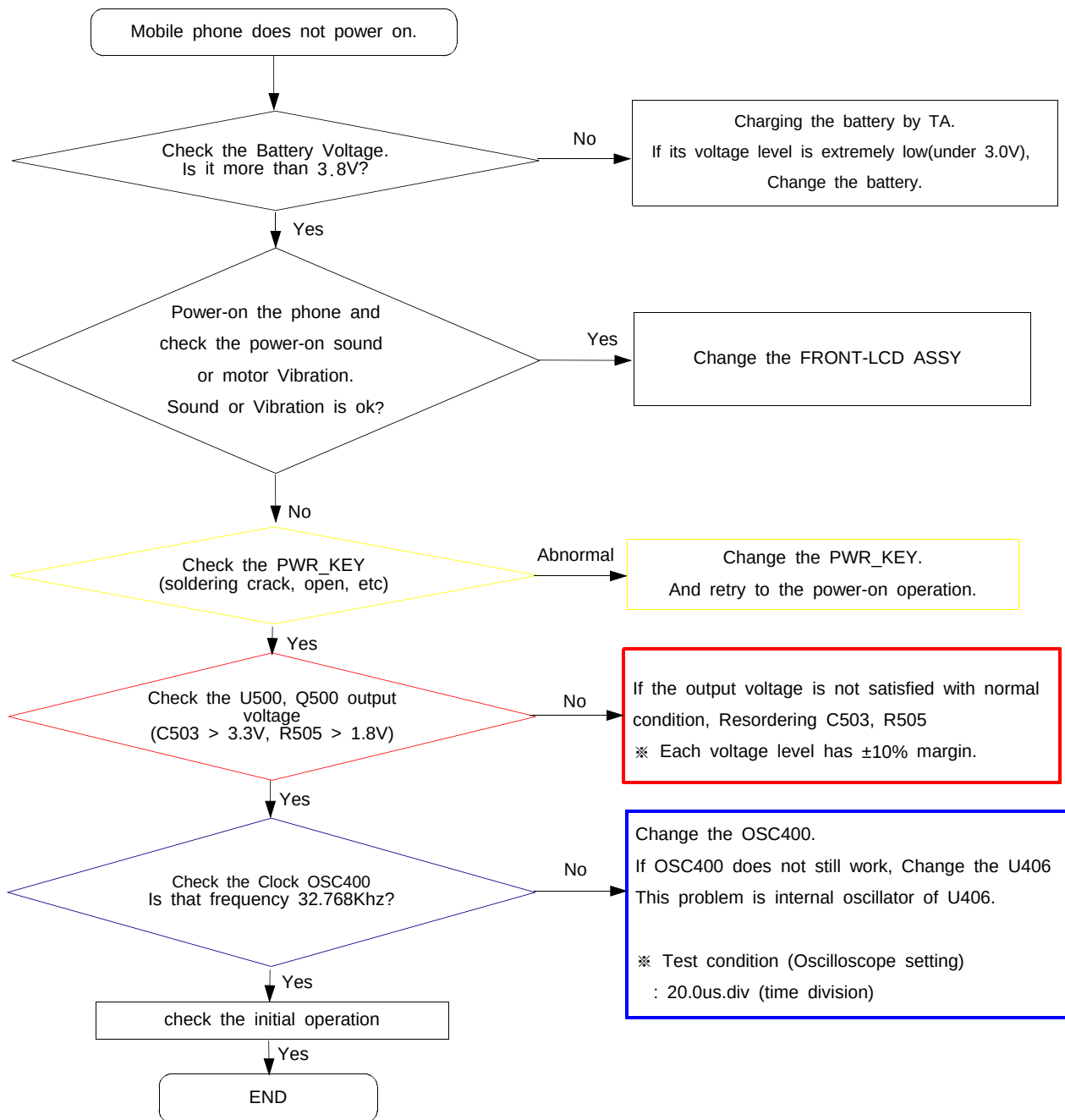


↑ 8960 & Spectrum Analyzer

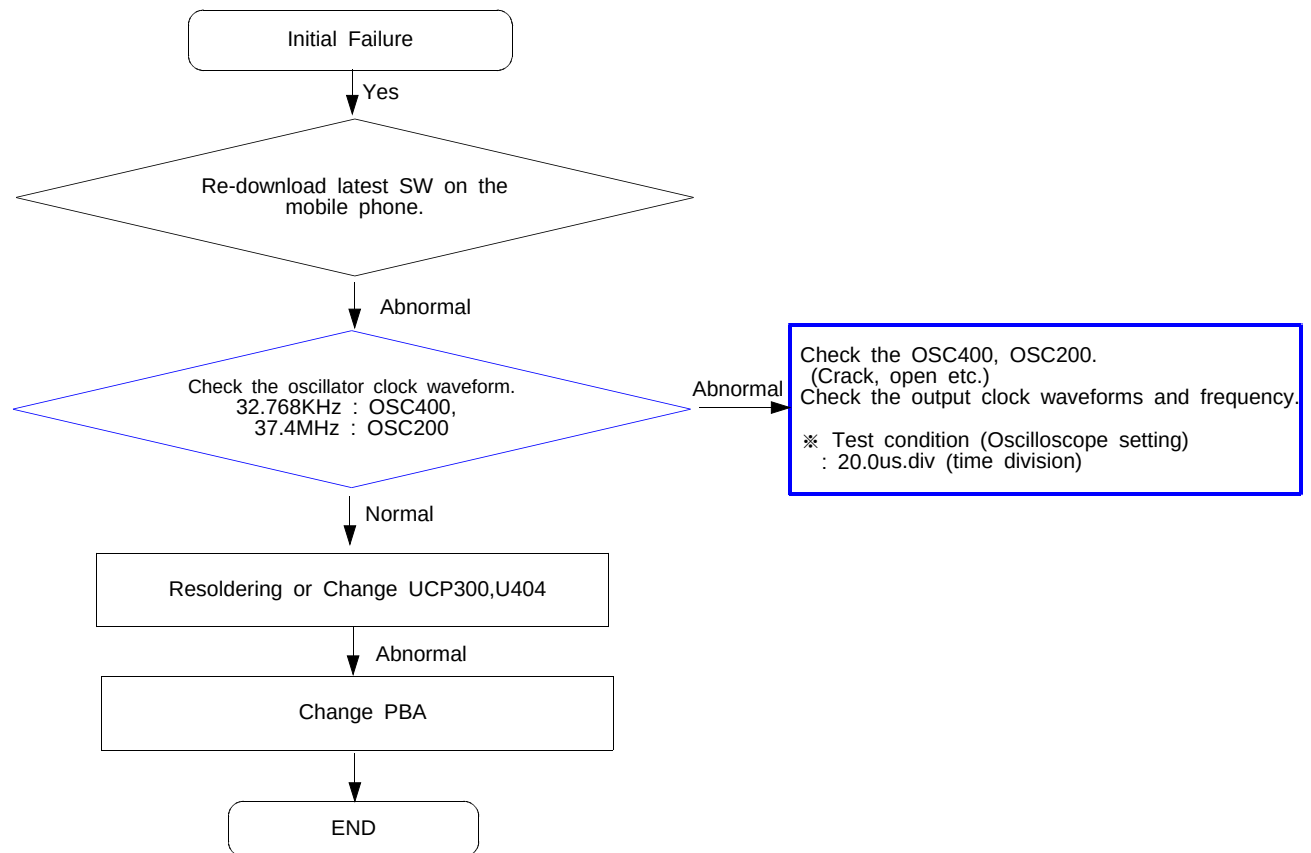


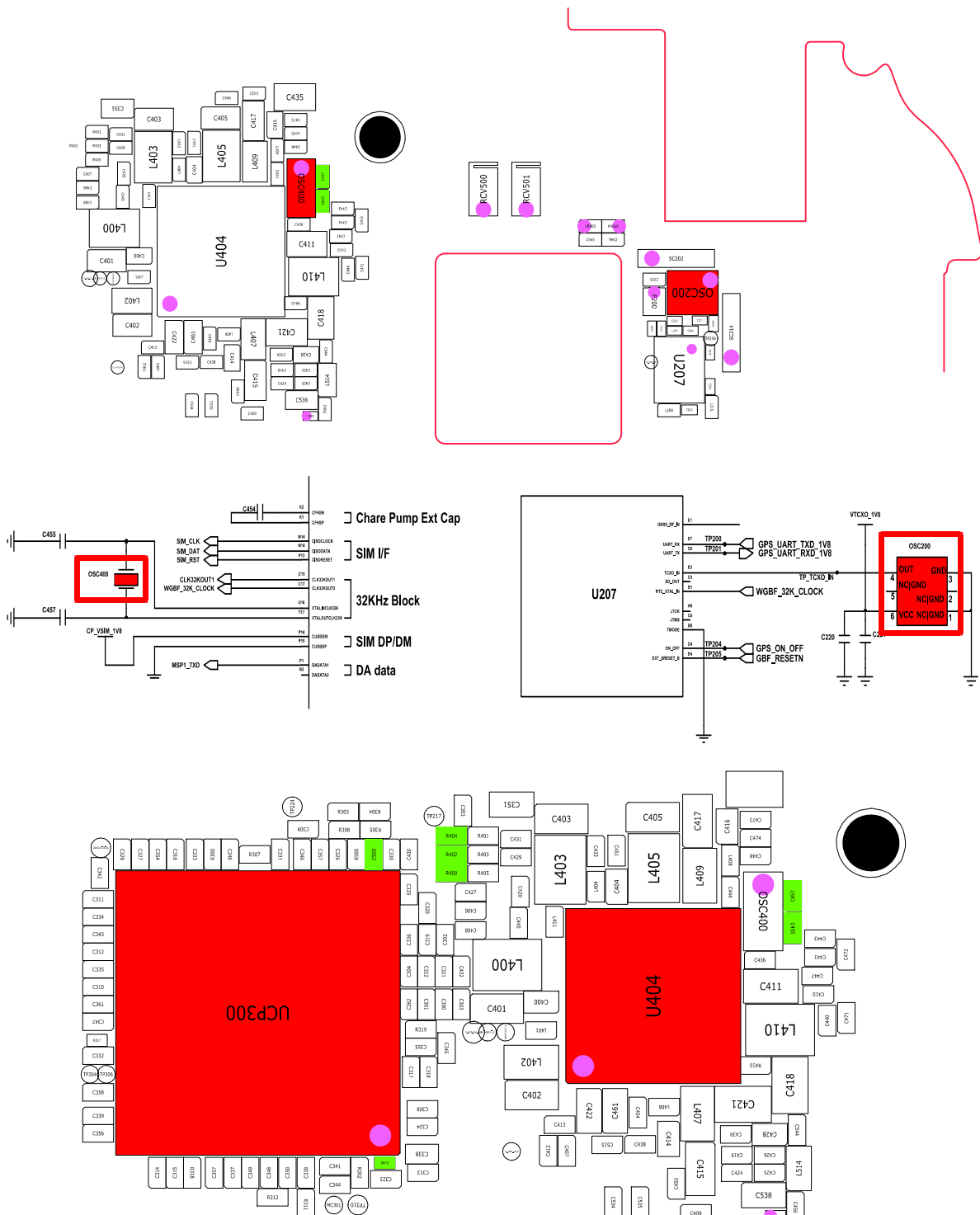
↗ Soldering iron

8-3-1. Power On

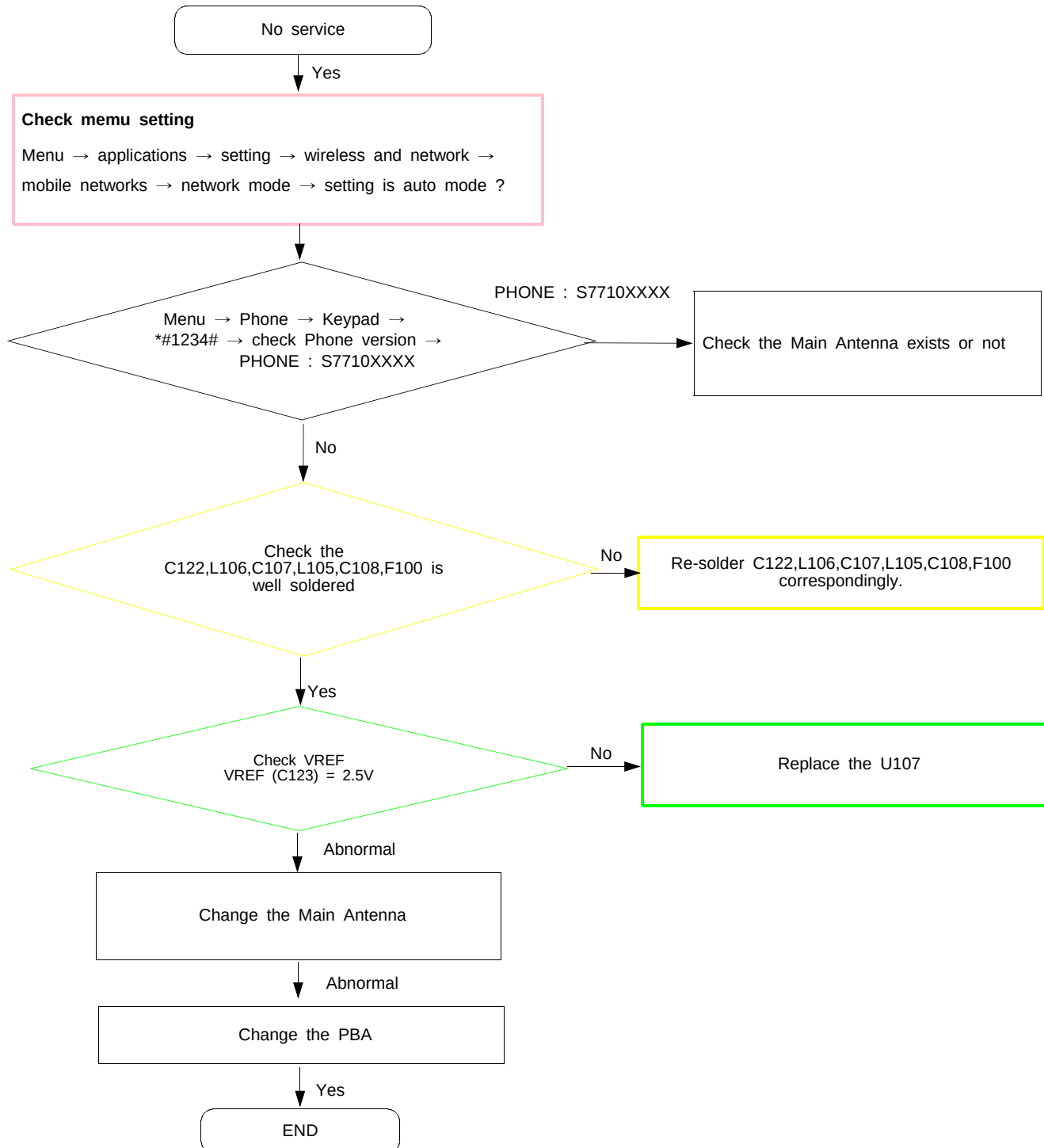


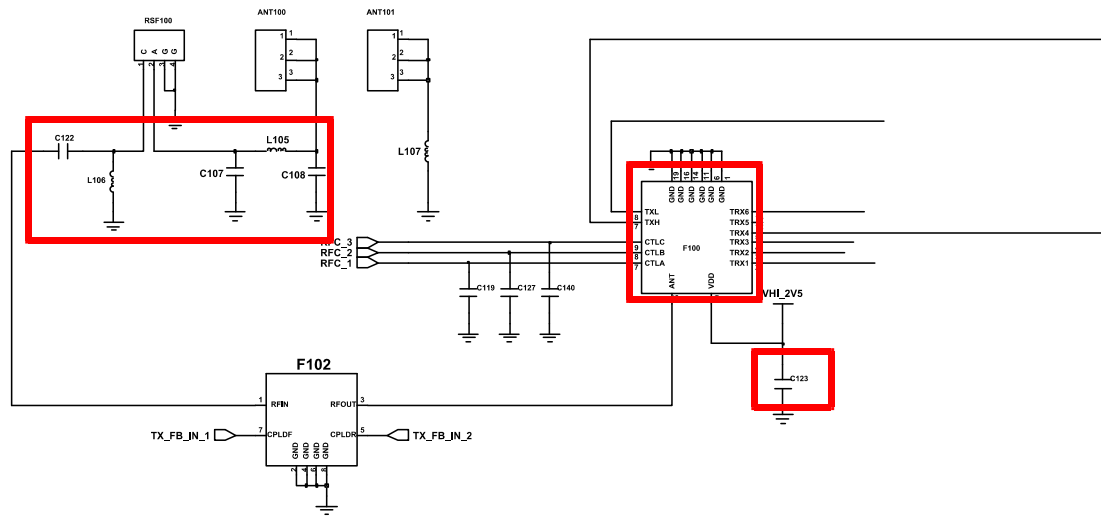
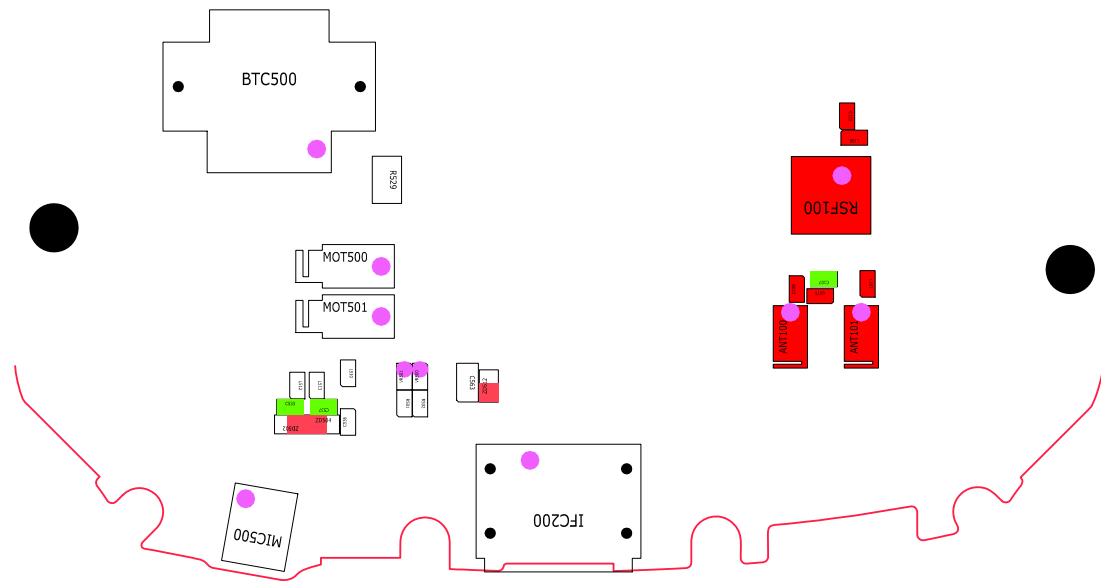
8-3-2. Initial



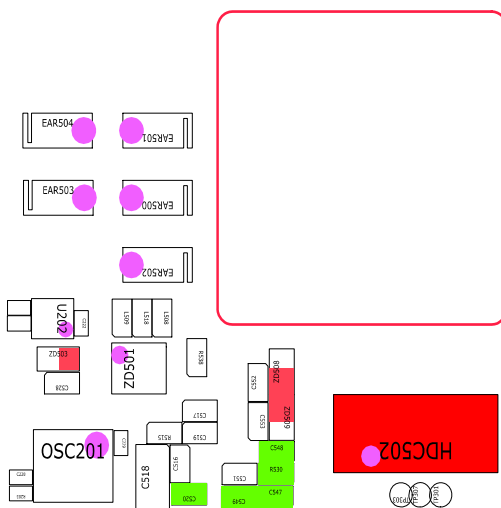
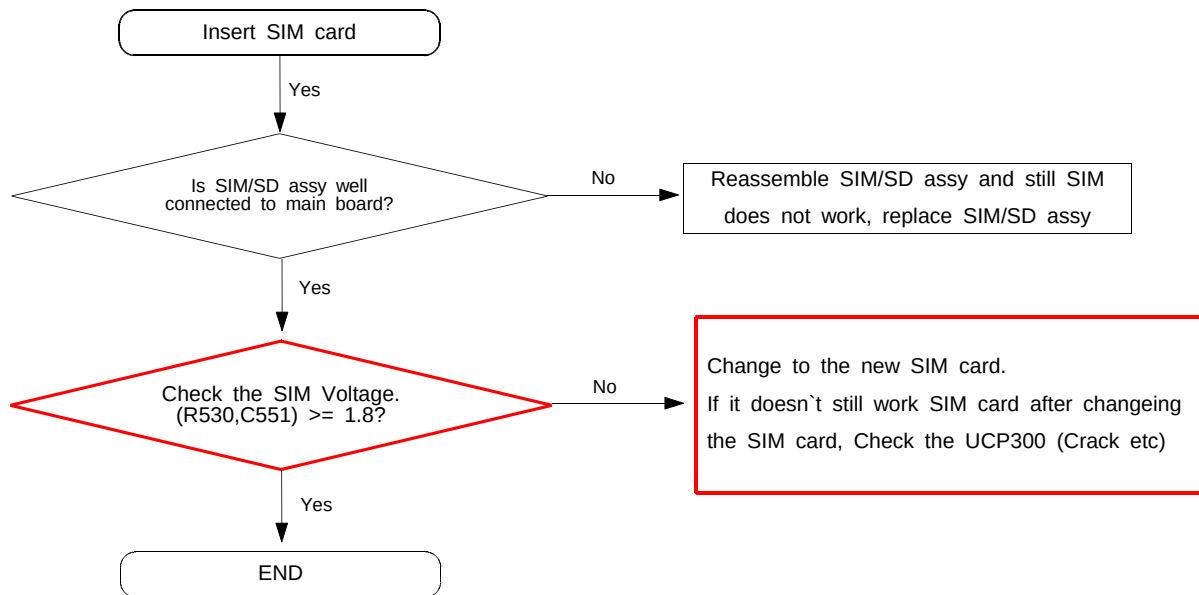


8-3-3. No Service

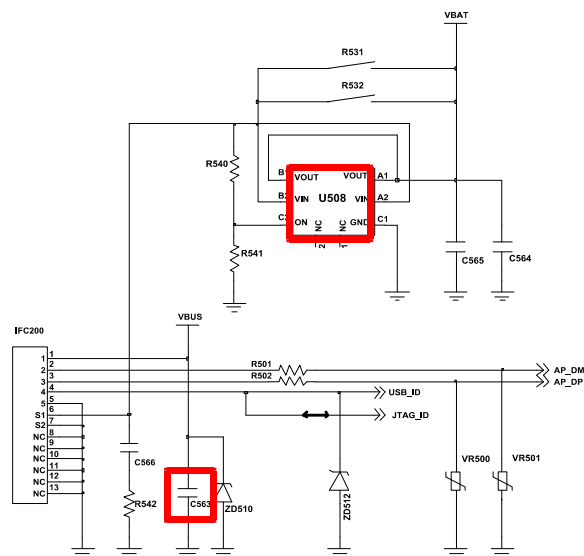
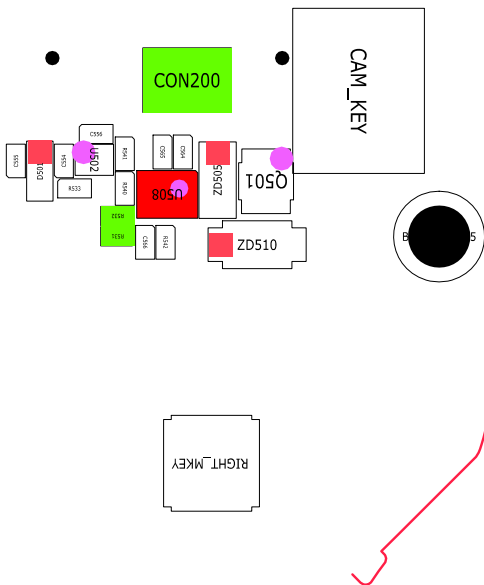
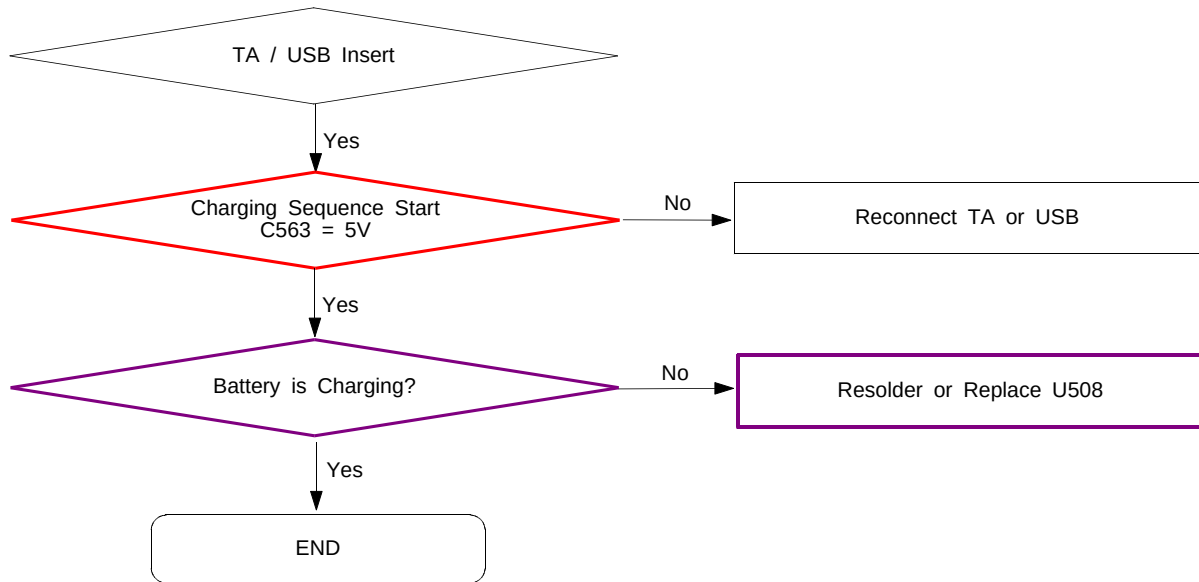




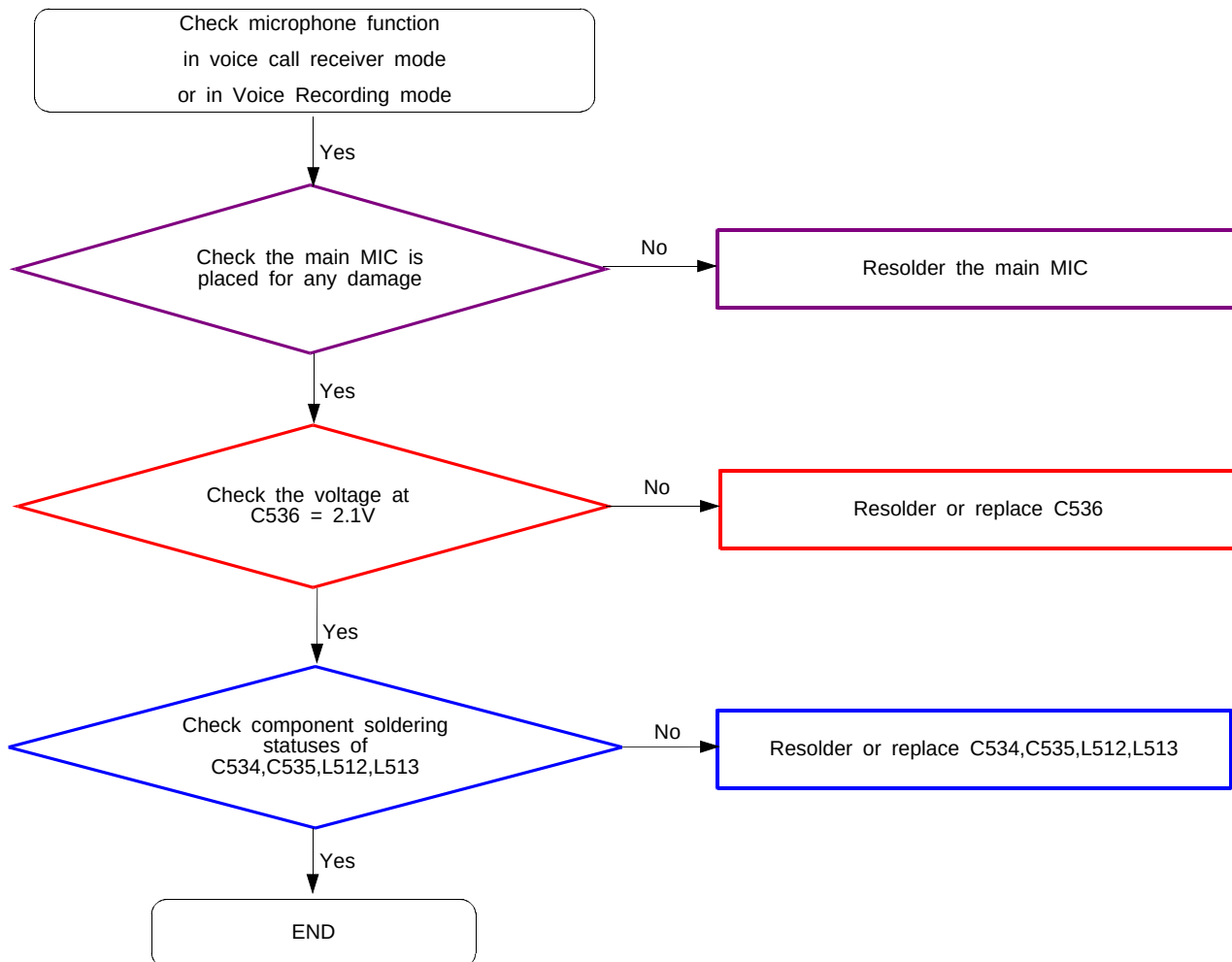
8-3-4. Sim Part

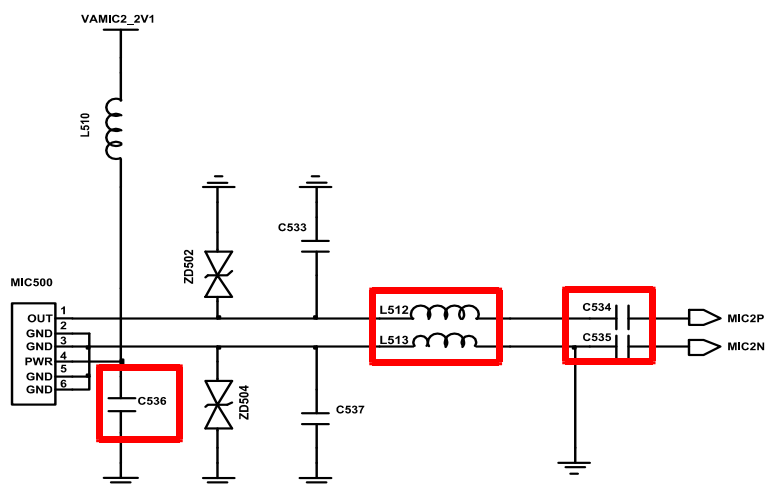
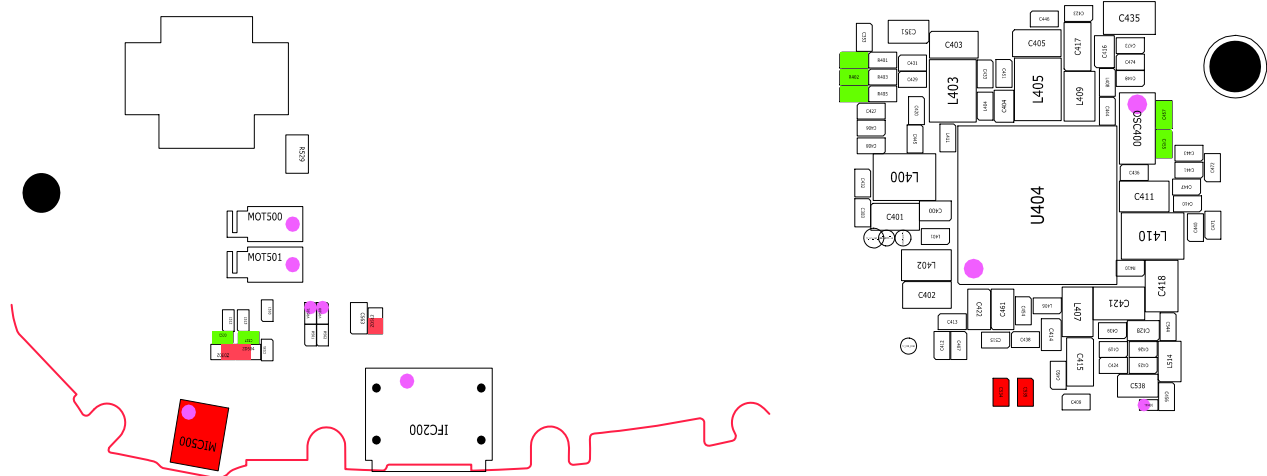


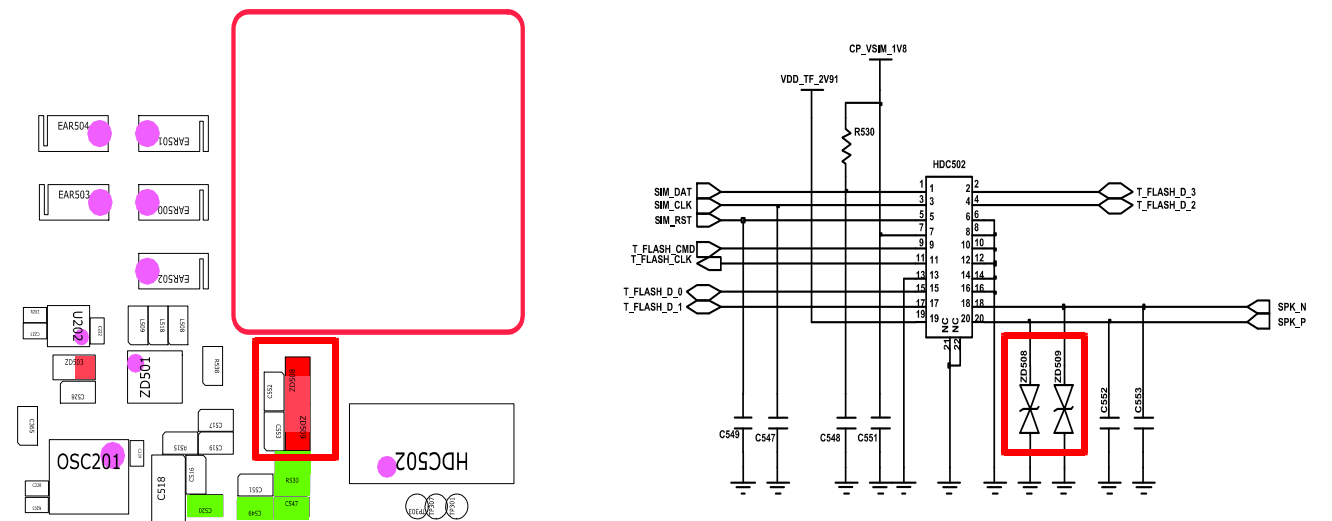
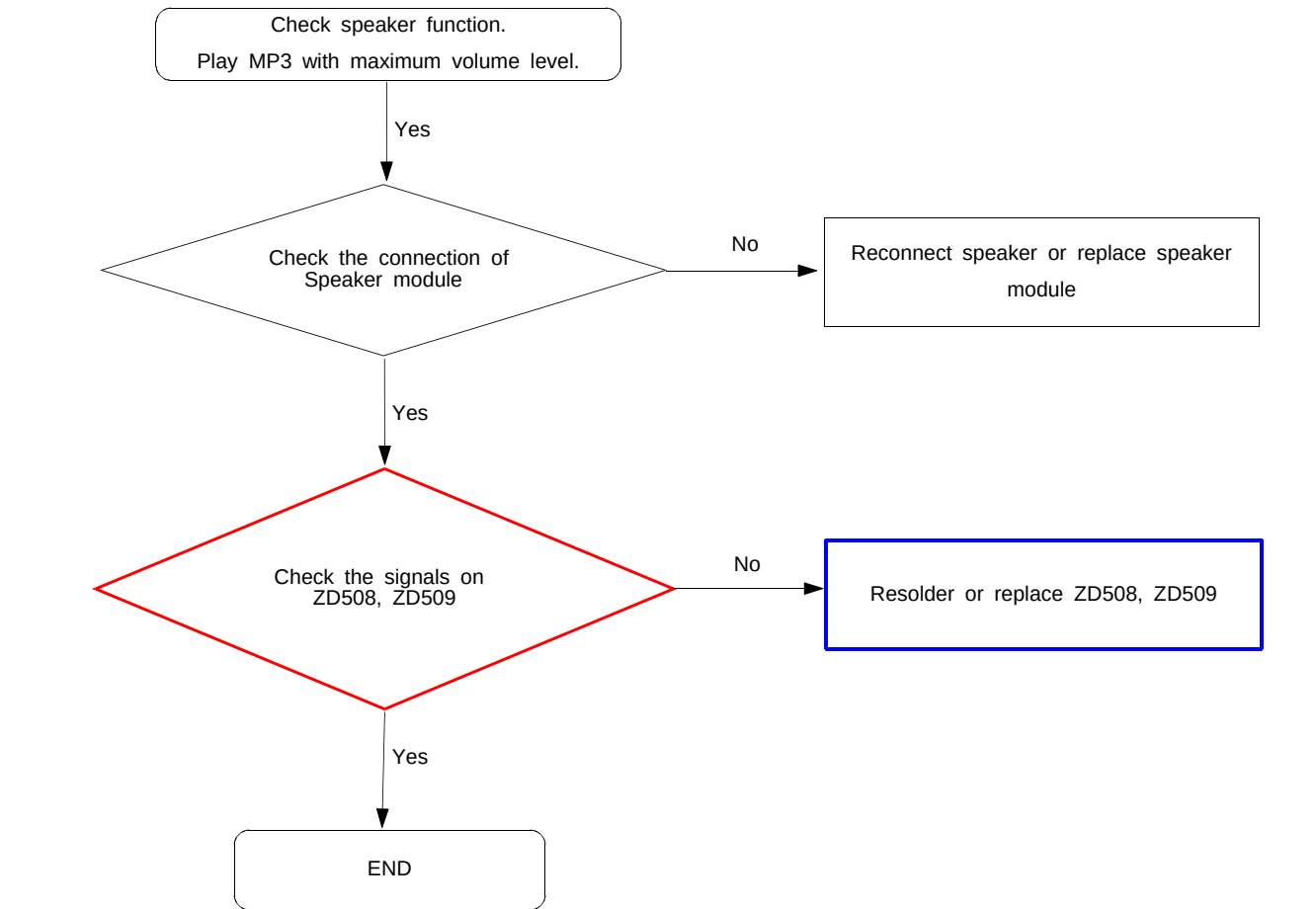
8-3-5. Charging Part



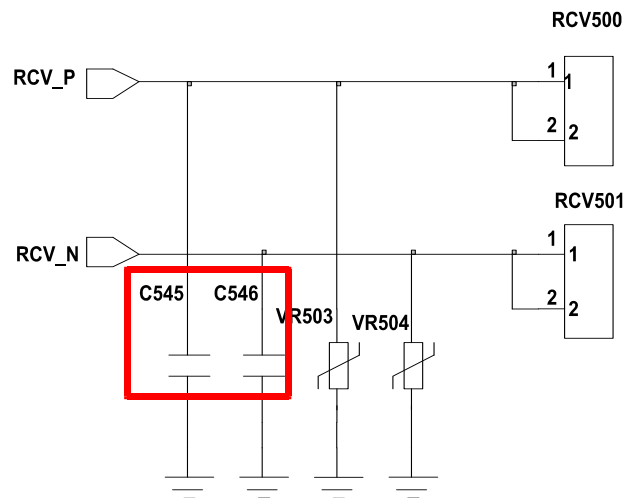
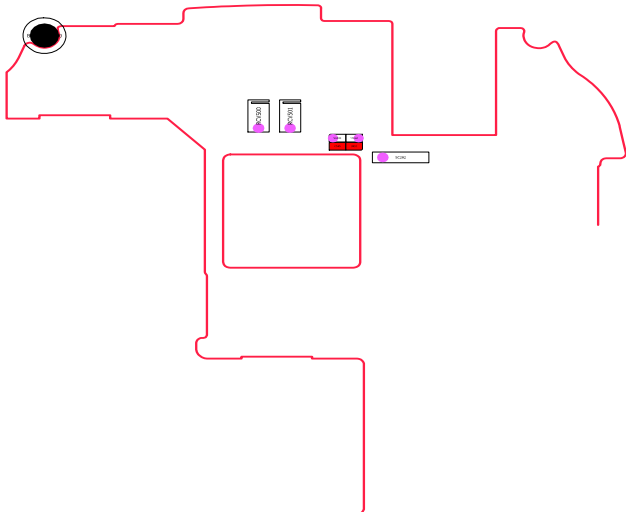
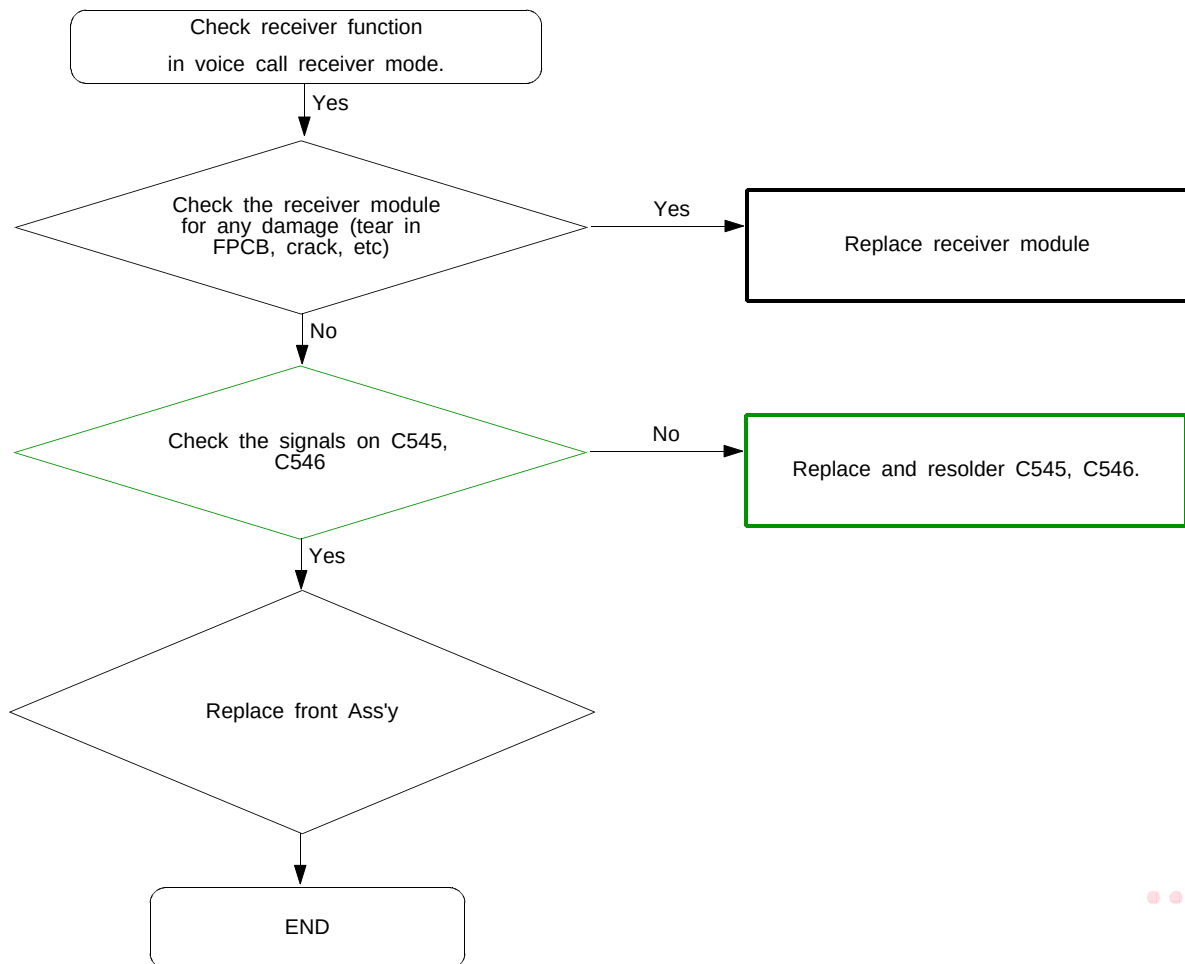
8-3-6. Microphone Part



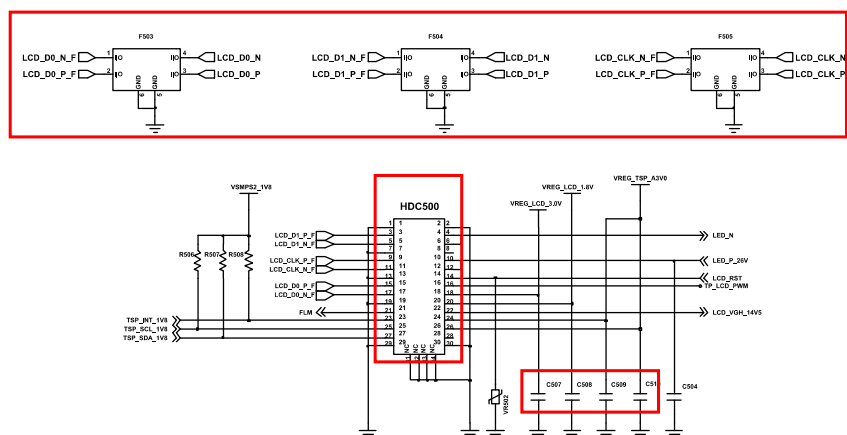
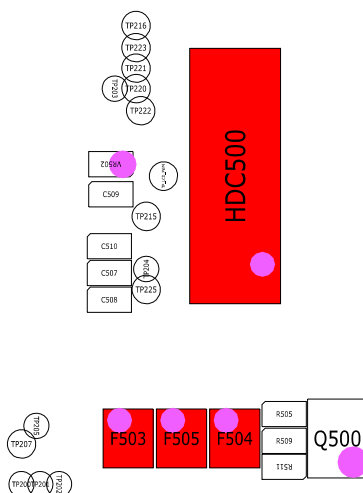
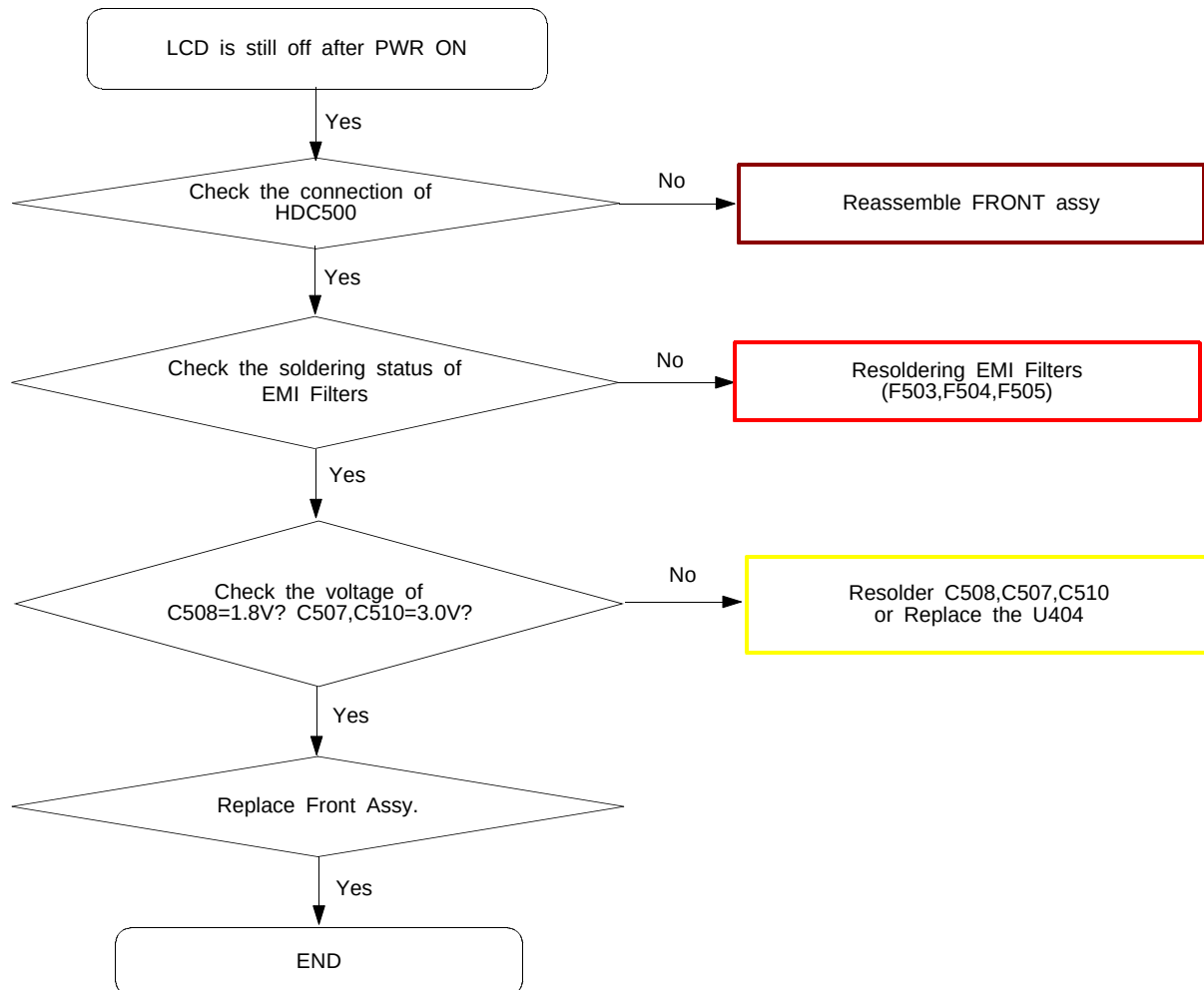




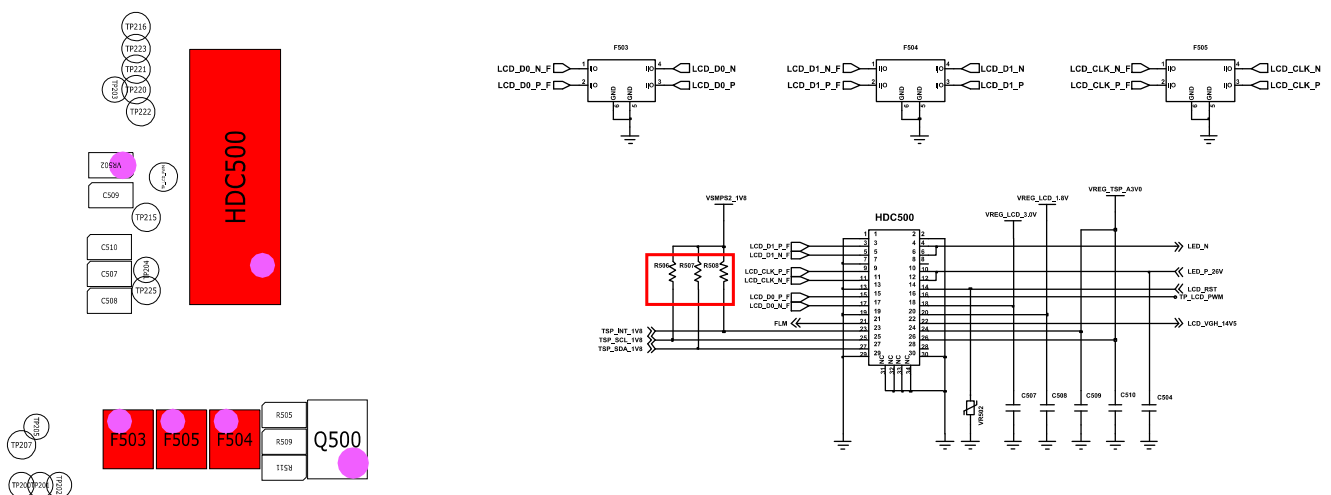
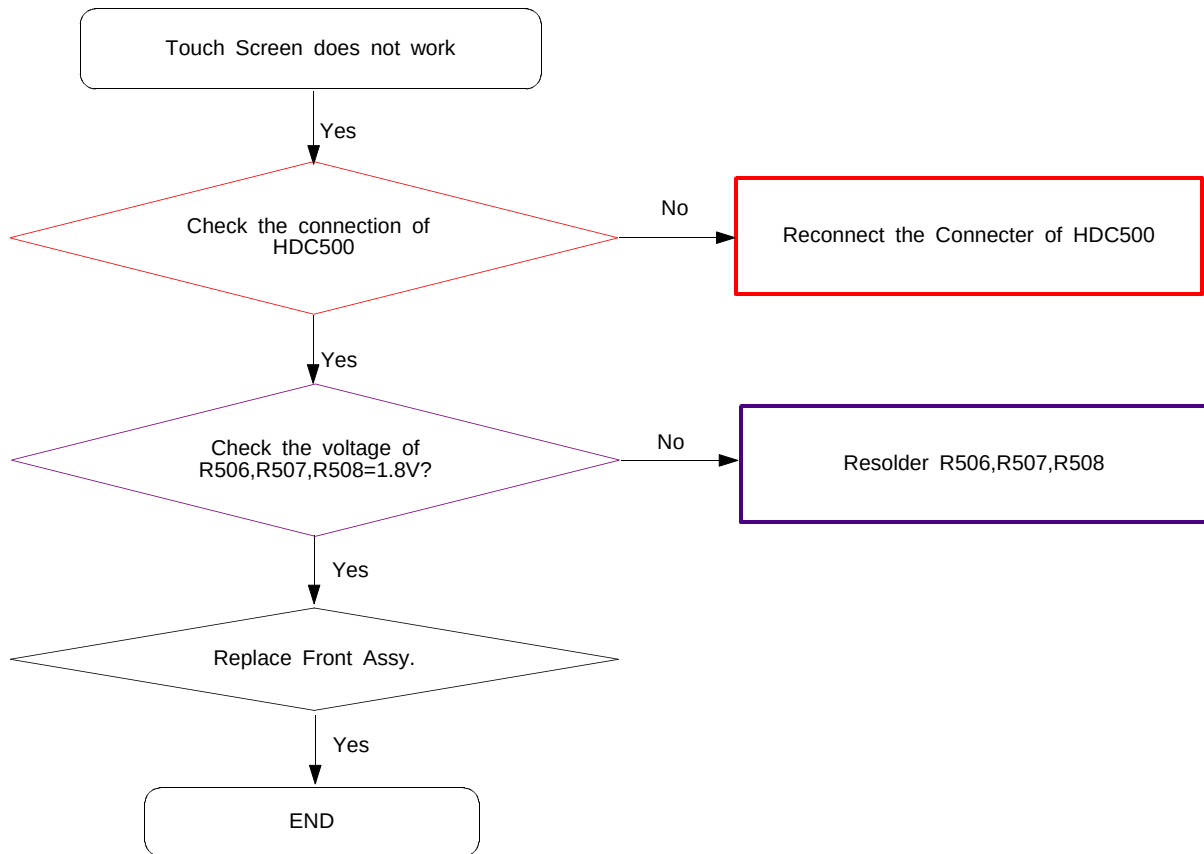
8-3-8 Receiver Part



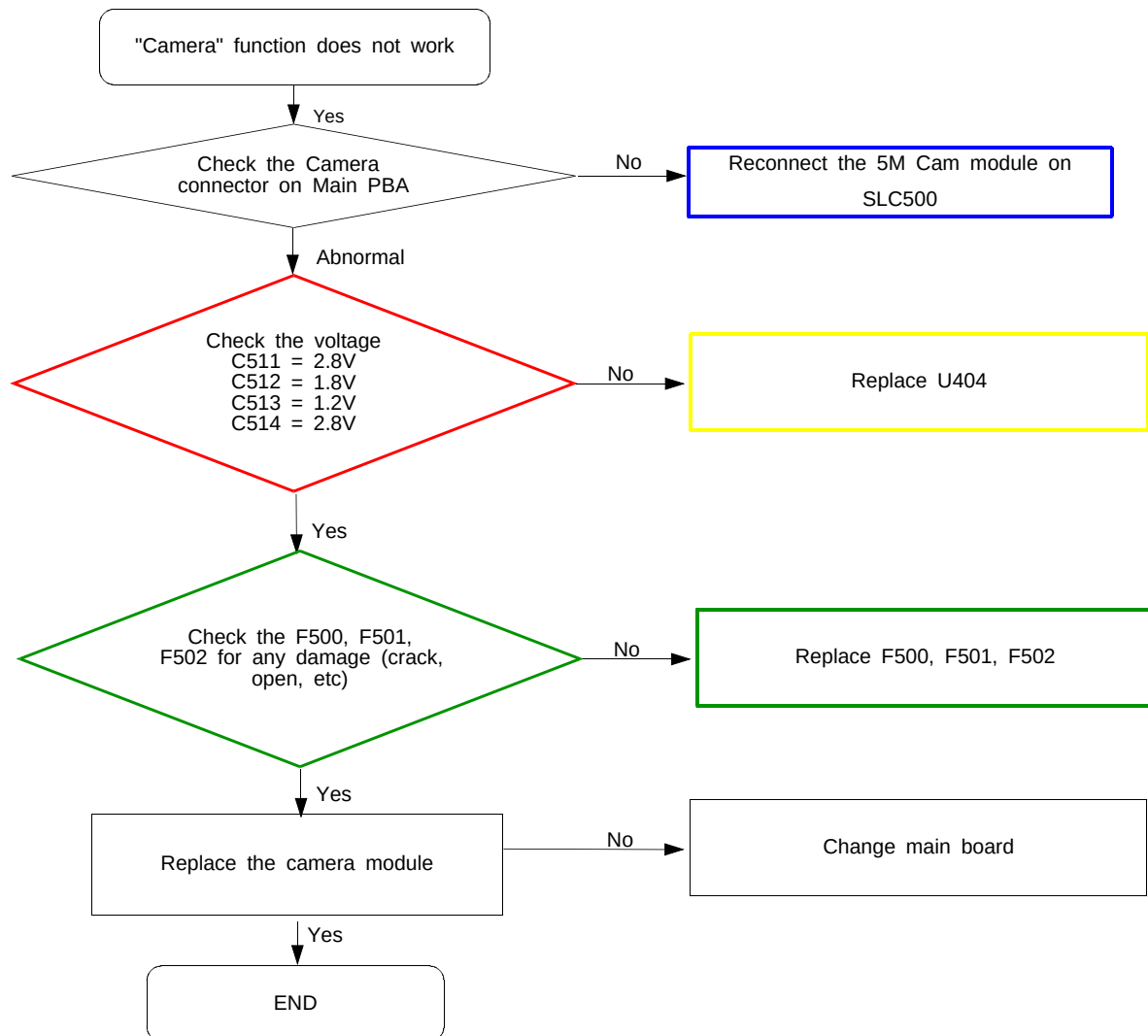
8-3-11. LCD



8-3-12. TSP



8-3-13. 5M CAM



※ presetting 8960

Call Setup Screen			
Control	Call Setup		Call Params
Operating Mode	DUT Information		BCH Parameters
Active Cell	IMSI: 001010123456789	Multislot Class (GPRS): ----	TCH Parameters
Connection Type	Called Num: 112	Multislot Class (EGPRS): ----	
Auto	Traffic Channel Downlink Power		POTCH Parameters
Originate Call	Burst 1, 2, 3, 4: ----, ----, ----, ---- dBm		
Paging IMSI	Counters		Receiver Control
001010123456789	Page: 0 DUT IP Tx, Packets: ----		
Handover Setup	PRACH: 0 DUT IP Rx, Bytes: ----		
Cell Info	Missing Burst: 0 DUT IP Rx, Packets: ----		
	Corrupt Burst: 0 DUT IP Rx, Bytes: ----		
	Decode Error: 0		
	Error Reports		
	Burst Timing Error: ---- T		
	BLER (Block Error Rate): ---- % over ---- blocks		
	USF BLER: ---- % over ---- blocks		
	Active Cell	Sys Type: GPRS	
	Idle		
1 of 2	Inlier Offset		

< 8960장비 초기화면 >

(Rx setting)

1. Active Cell : select GSM or GPRS
2. Connection Type : select Auto(GSM), BLER(GPRS)
3. BCH Parameter : select measuring band (DCS or EGSM)
4. Cell power : -60dBm

Call Setup Screen			
Control	Call Setup		Call Params
Operating Mode	DUT Information for IMEI 010945000000650		BCH Parameters
Active Cell	IMSI: 001010123456789	Multislot Class (GPRS): ----	TCH Parameters
Connection Type	Called Num: 112	Multislot Class (EGPRS): ----	
Auto	Traffic Channel Downlink Power		POTCH Parameters
End Call	Burst 1, 2, 3, 4: -60.00, ----, ----, ---- dBm		
Originate Call	Unused Bursts: ---- dBm		Receiver Control
Paging IMSI	Counters		
001010123456789	Page: 2 DUT IP Tx, Packets: ----		
Handover Setup	PRACH: 1 DUT IP Rx, Bytes: ----		
Cell Info	PRACH: 0 DUT IP Rx, Packets: ----		
	Missing Burst: 0 DUT IP Rx, Bytes: ----		
	Corrupt Burst: 0		
	Decode Error: 0		
	Error Reports		
	Burst Timing Error: 0.50 T		
	BLER (Block Error Rate): ---- % over ---- blocks		
	USF BLER: ---- % over ---- blocks		
	Active Cell	Sys Type: GSM	
	Connected		
1 of 2	Inlier Offset		

< Call이 연결된 화면 >

(Tx setting)

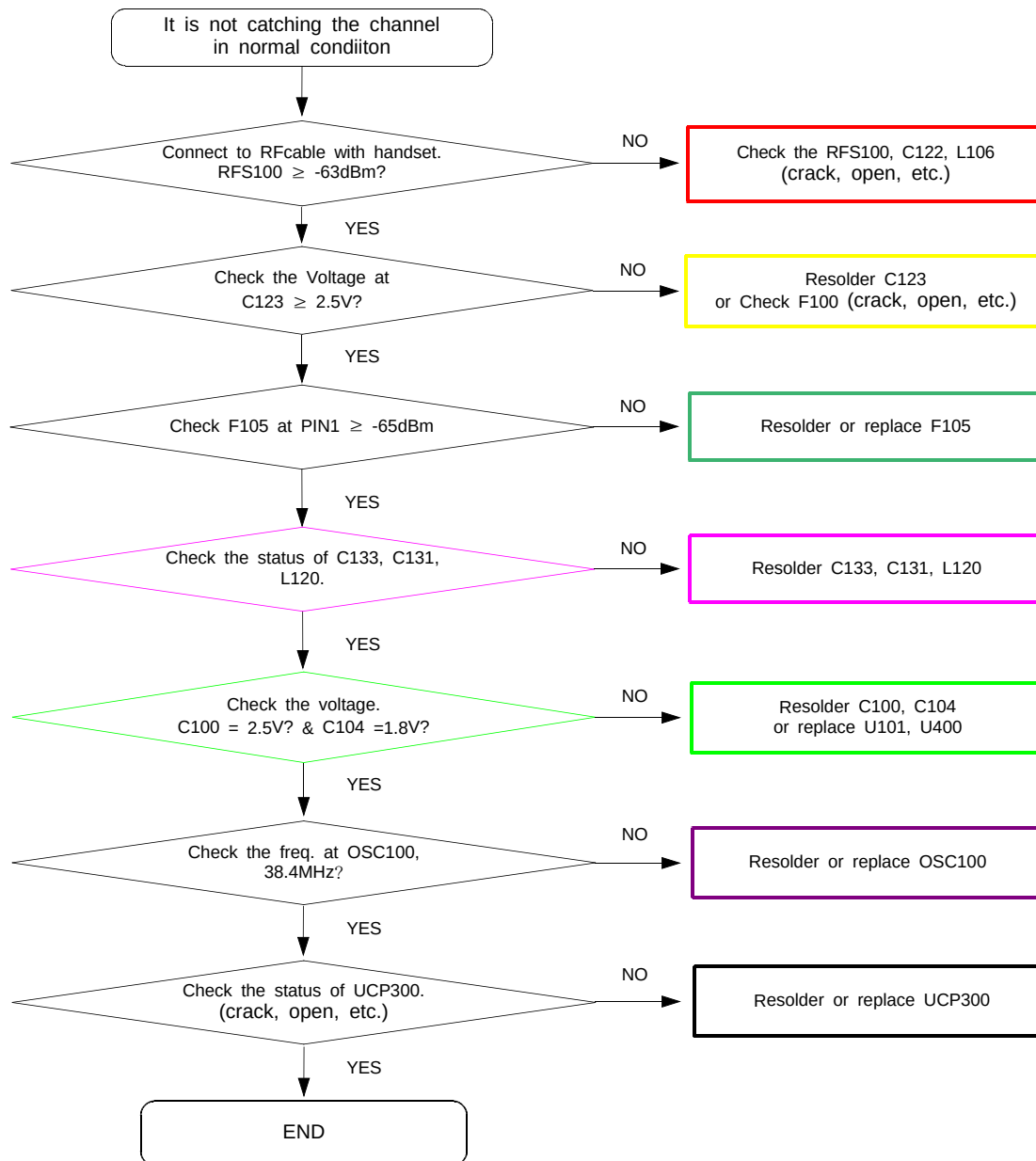
※ After setting 8960 (EGSM / DCS)

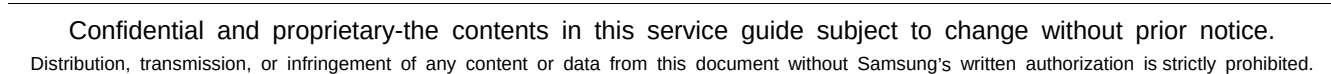
1. After setting, prepare the call setup Display
2. Using an Originate Call, make a call.
3. Confirm the display "connected"
4. start the measuring

※ 8960 & spectrum analyzer (down & up at picture)

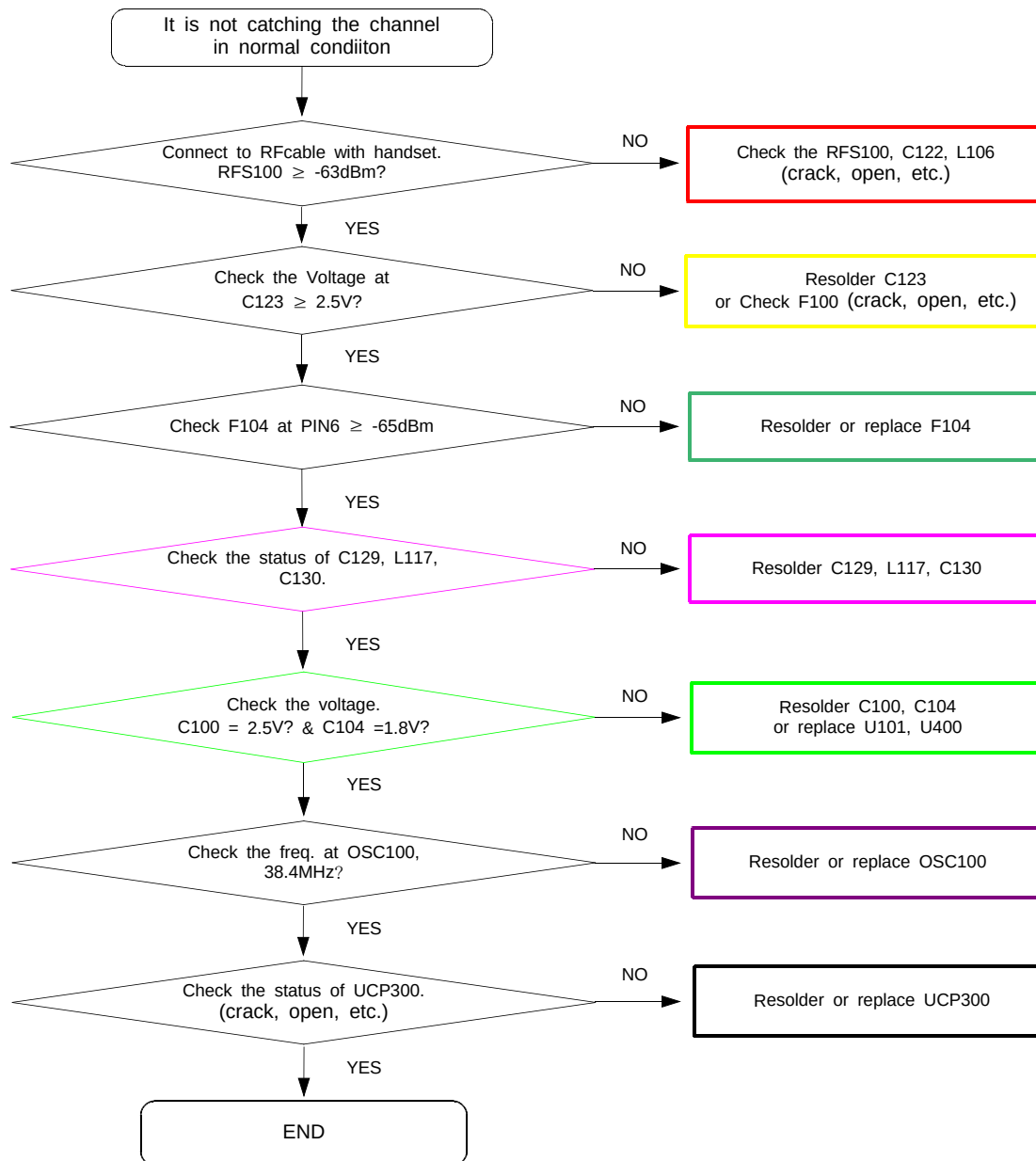
- spectrum analyzer : testing method = the way using an Oscilloscope
- 8960 : connect using RF Cable between 8960 & RF Connector in board.

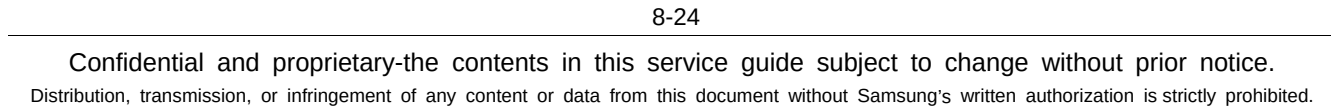
8-3-15. GSM850 Rx



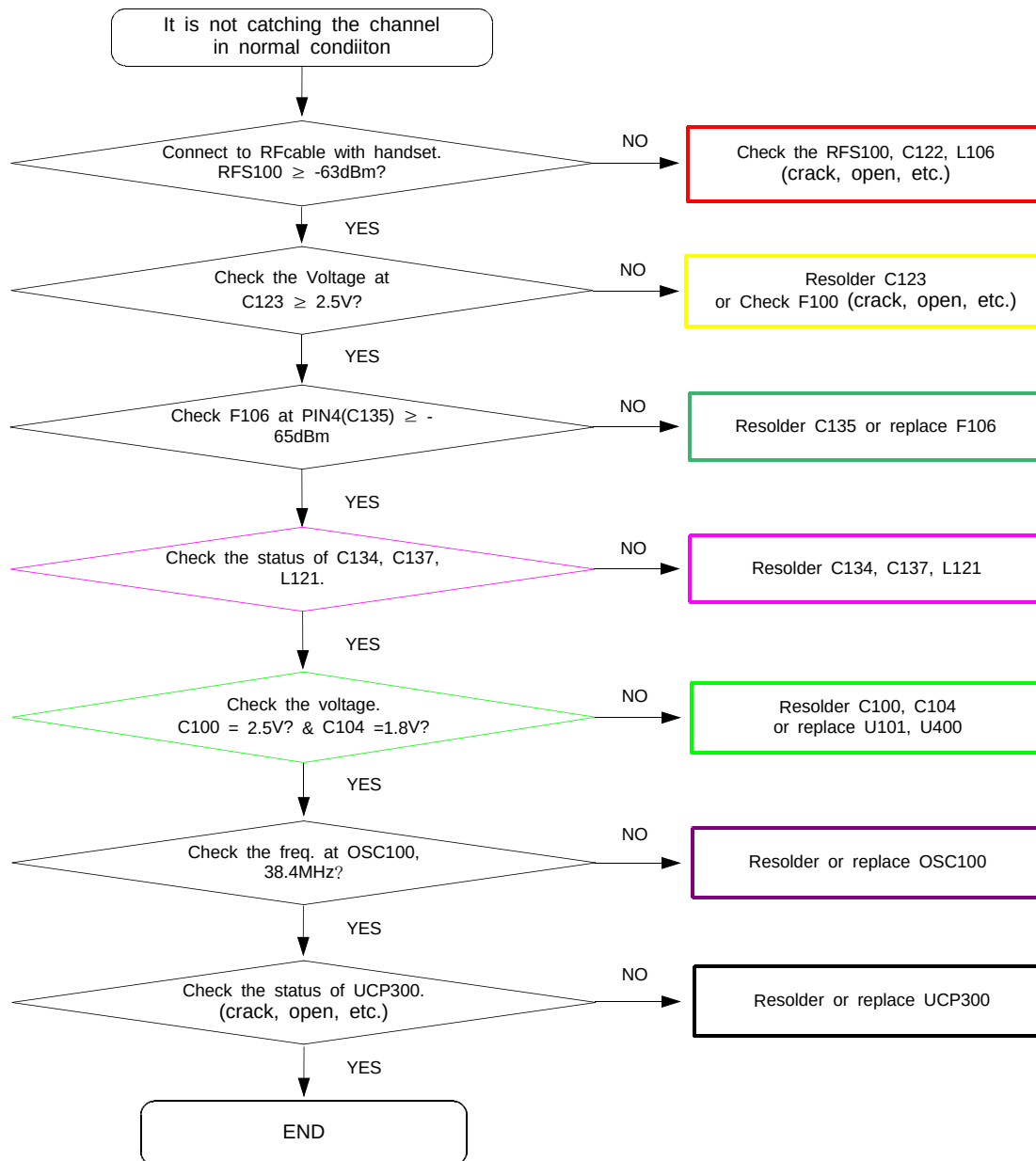


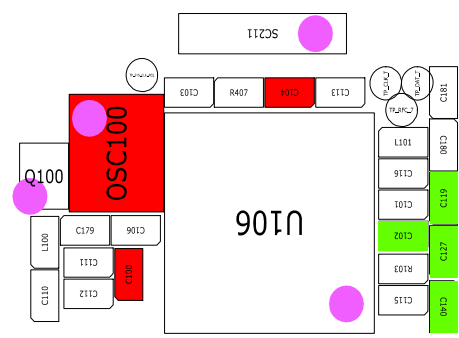
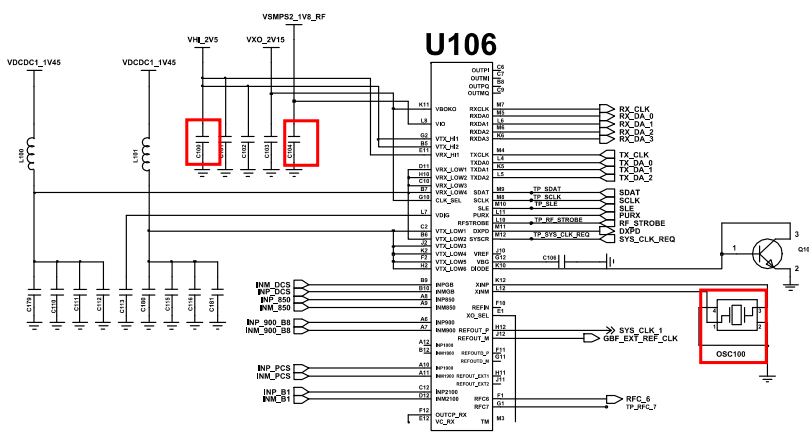
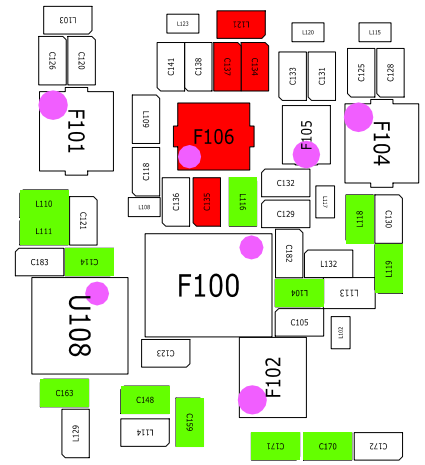
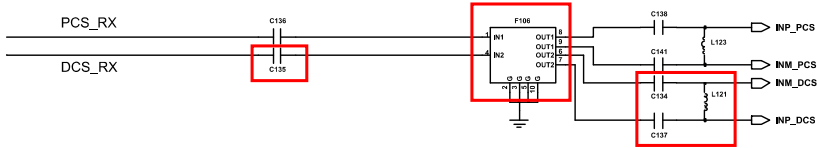
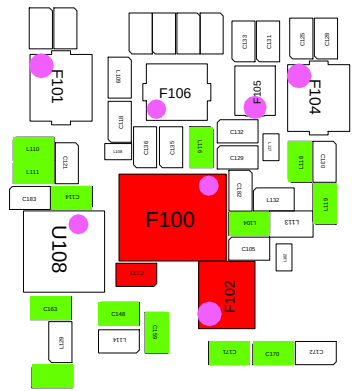
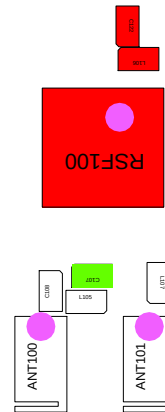
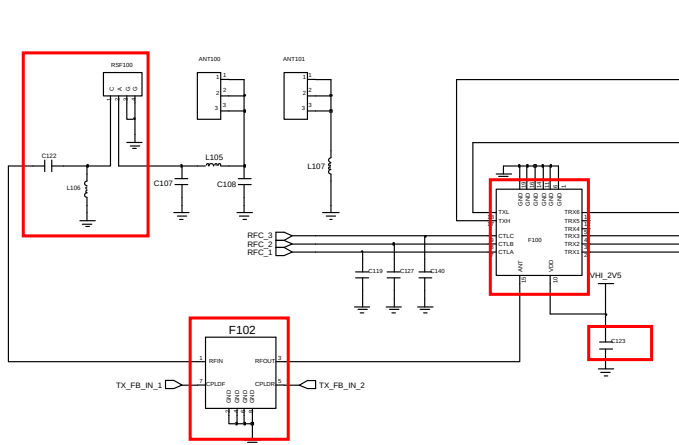
8-3-15. GSM 900 / WCDMA BAND 8 (900) Rx



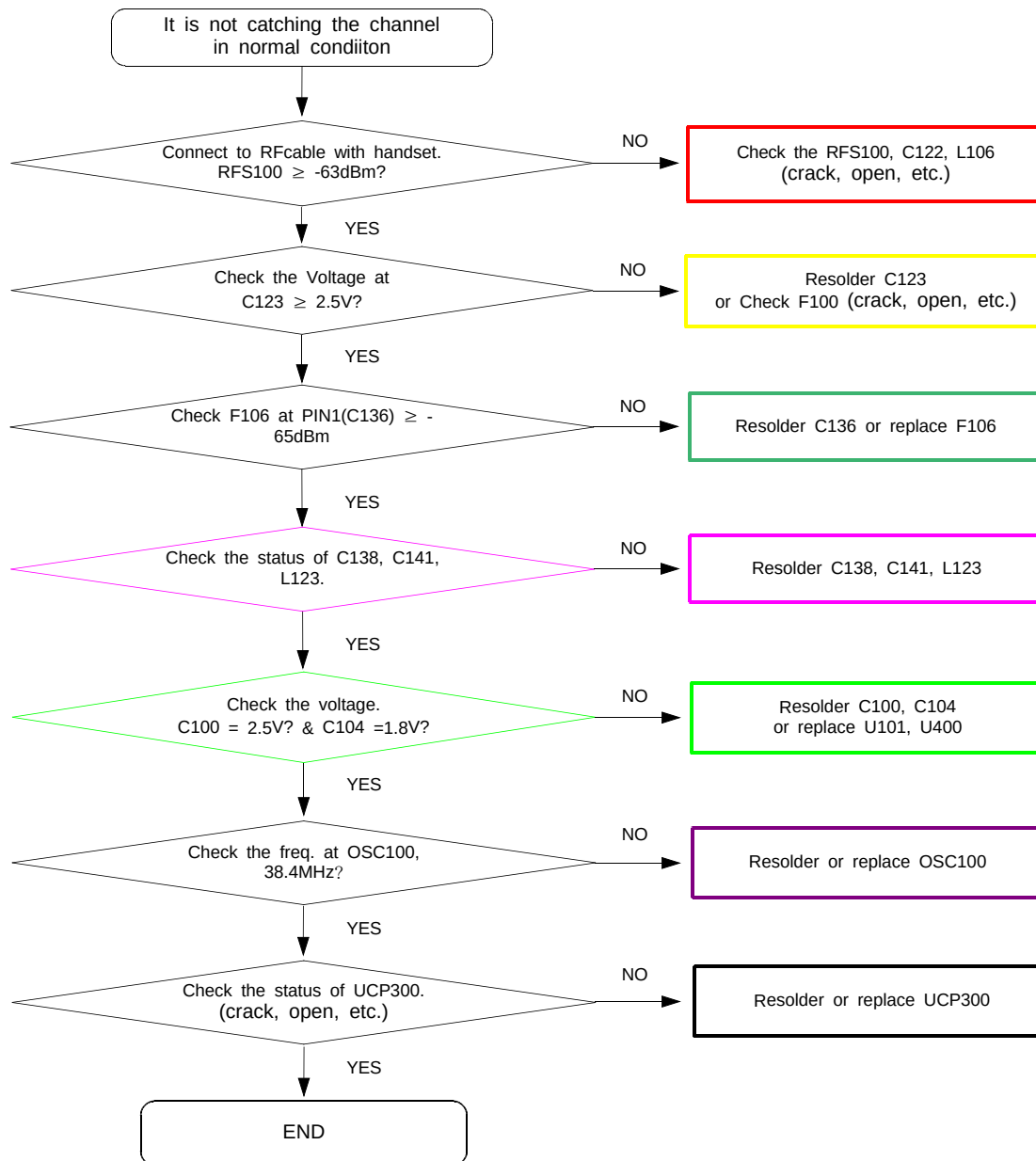


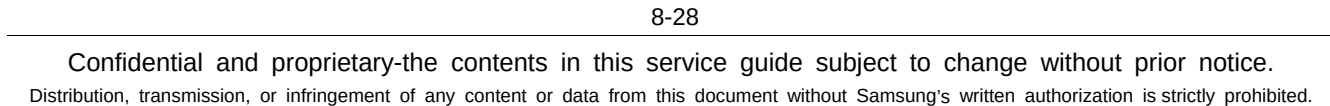
8-3-16. GSM1800 Rx



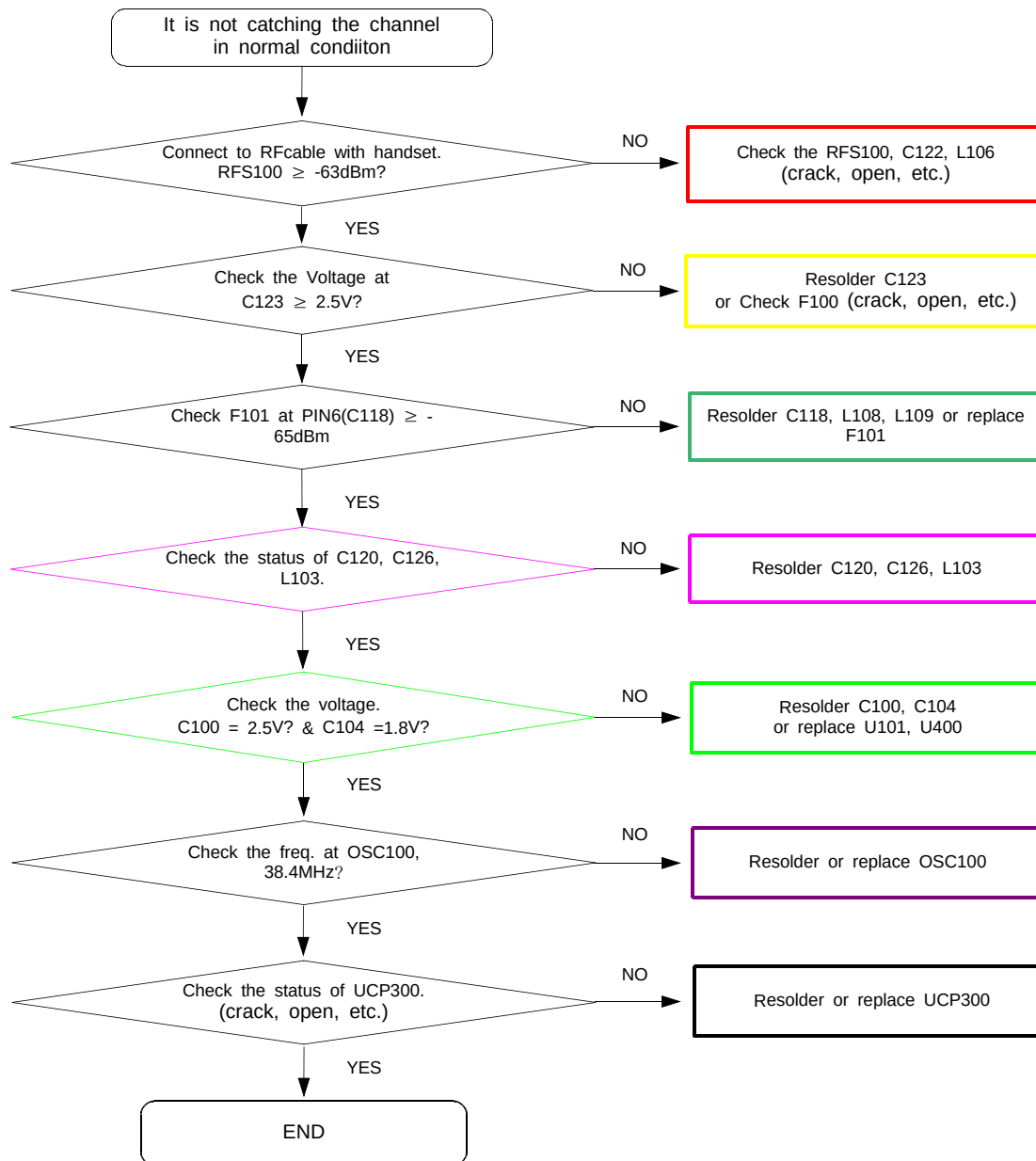


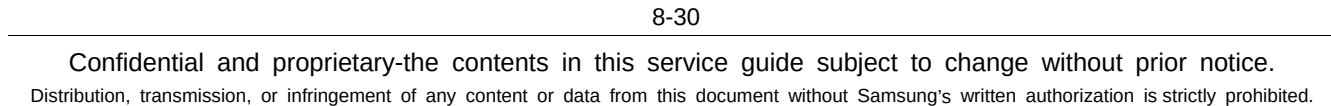
8-3-15. GSM1900 Rx



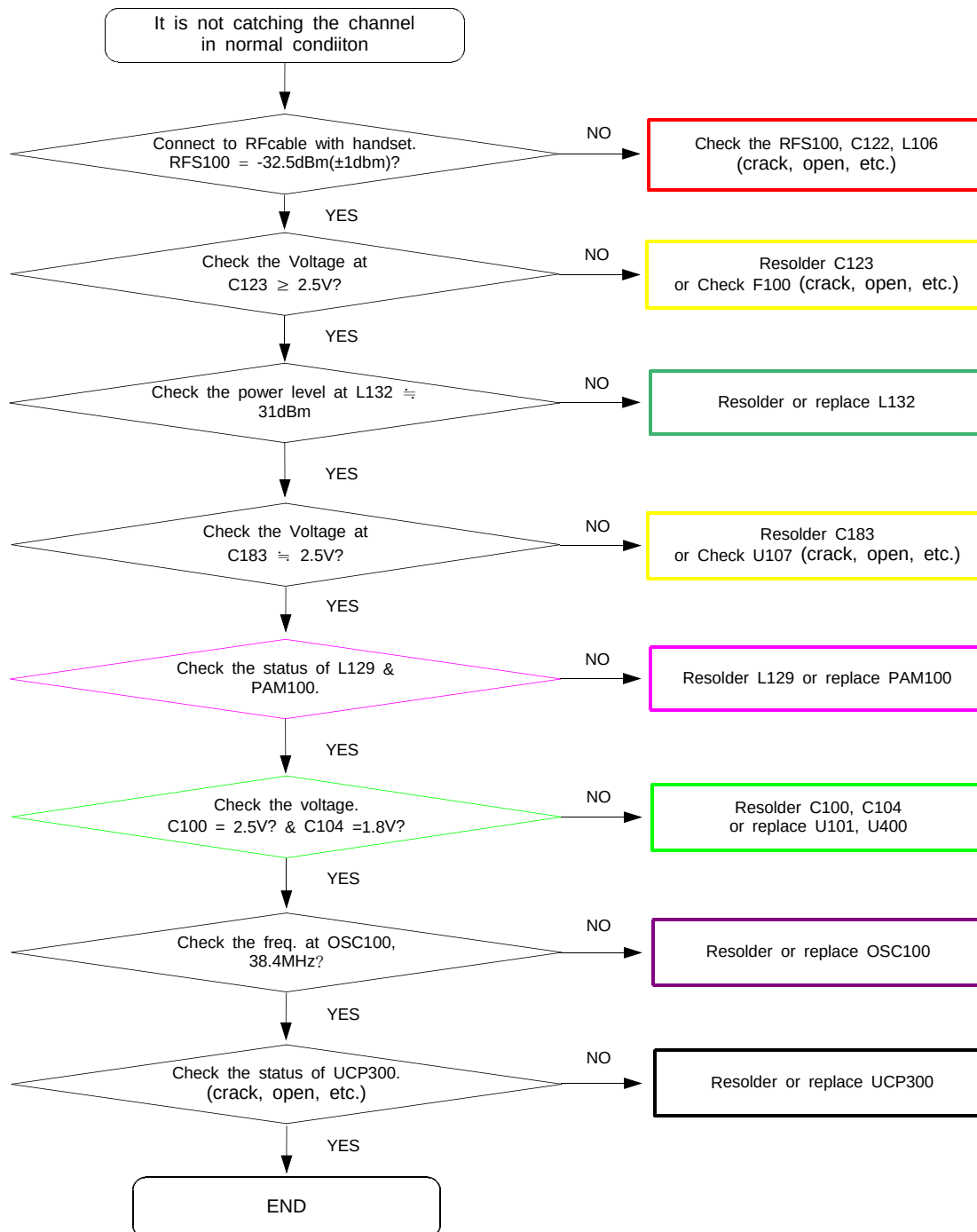


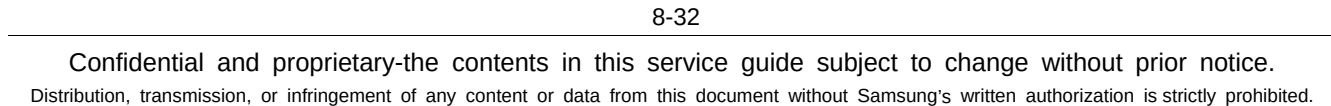
8-3-17. WCDMA Band1 Rx



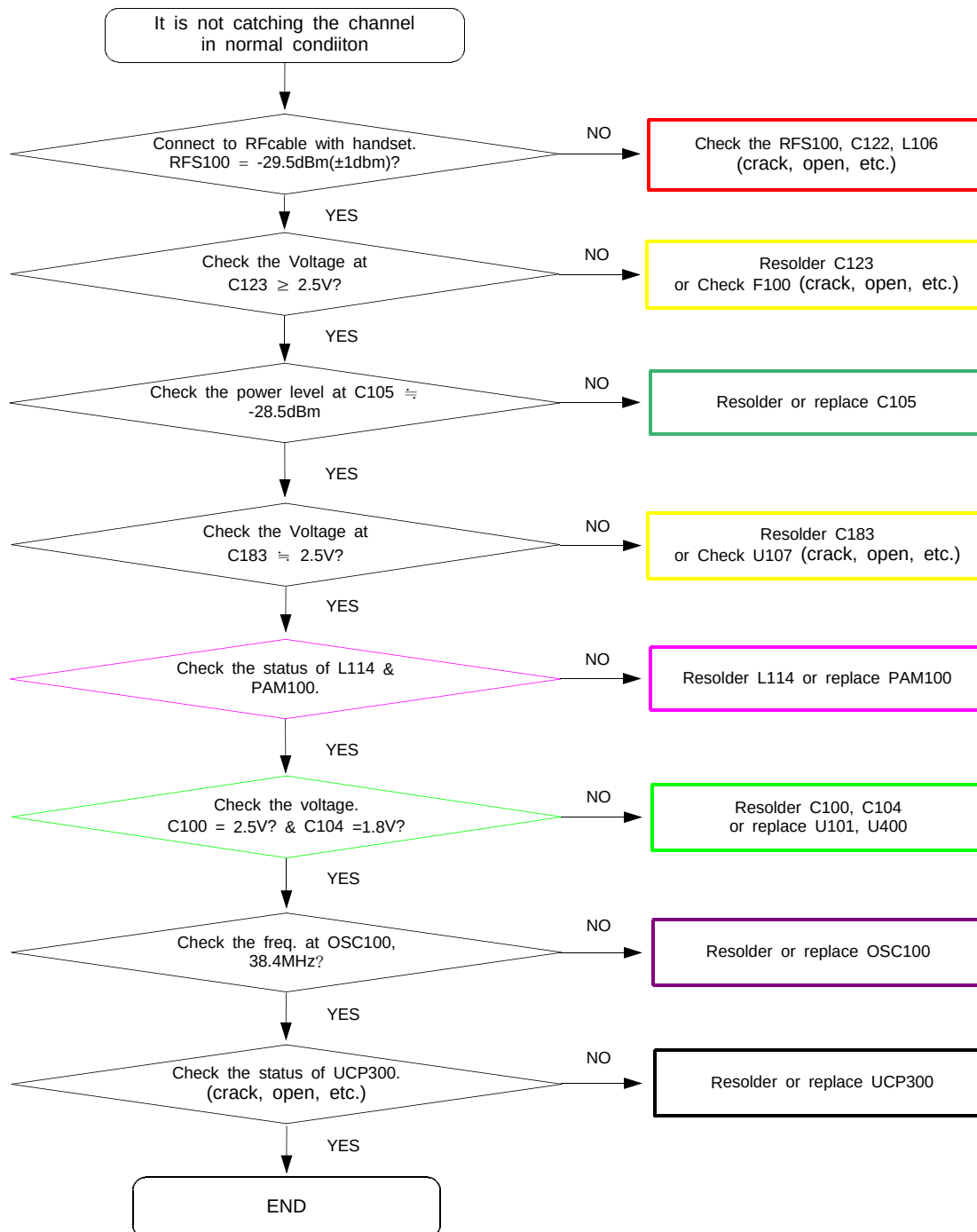


8-3-20. GSM850/900 Tx



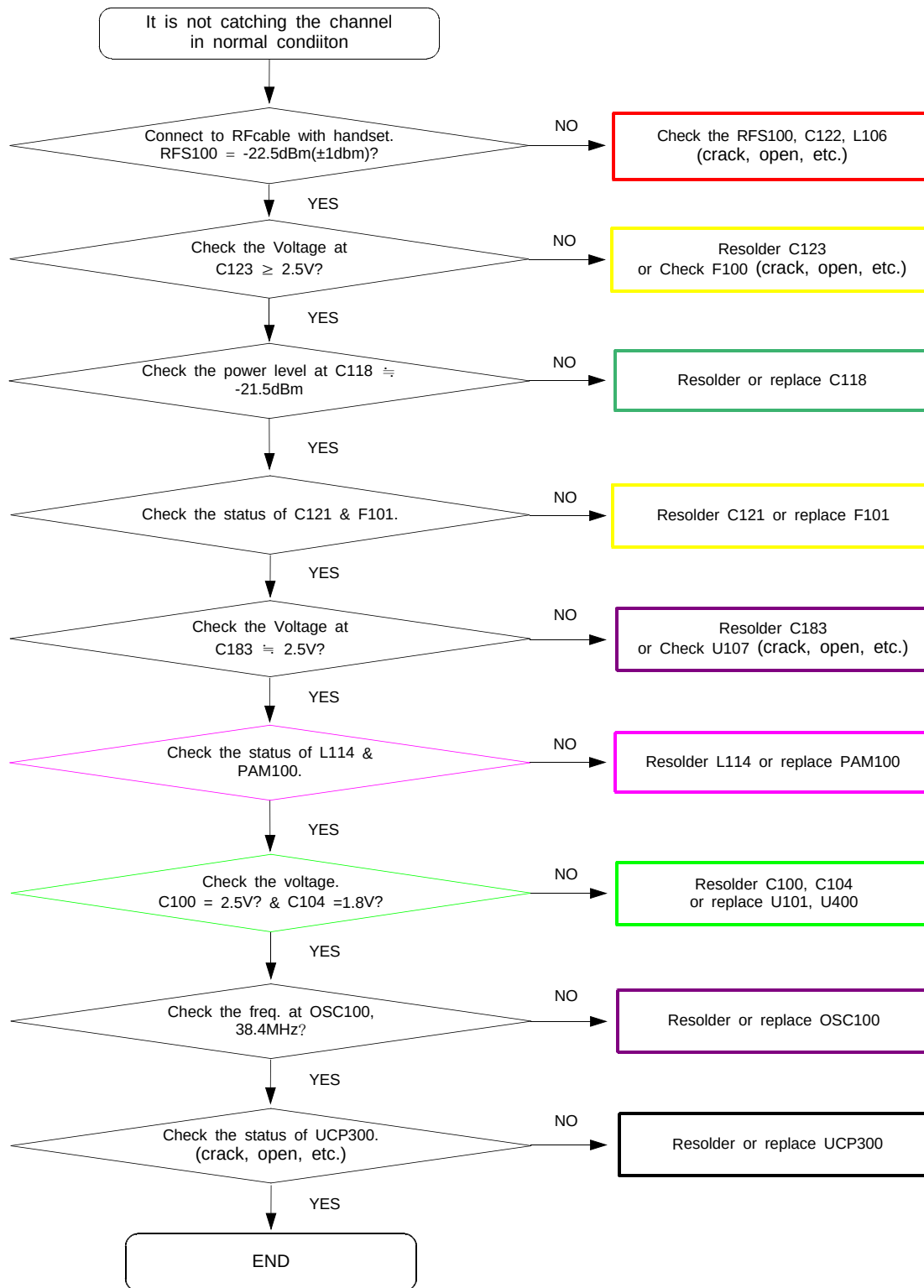


8-3-21. DCS/PCS Tx



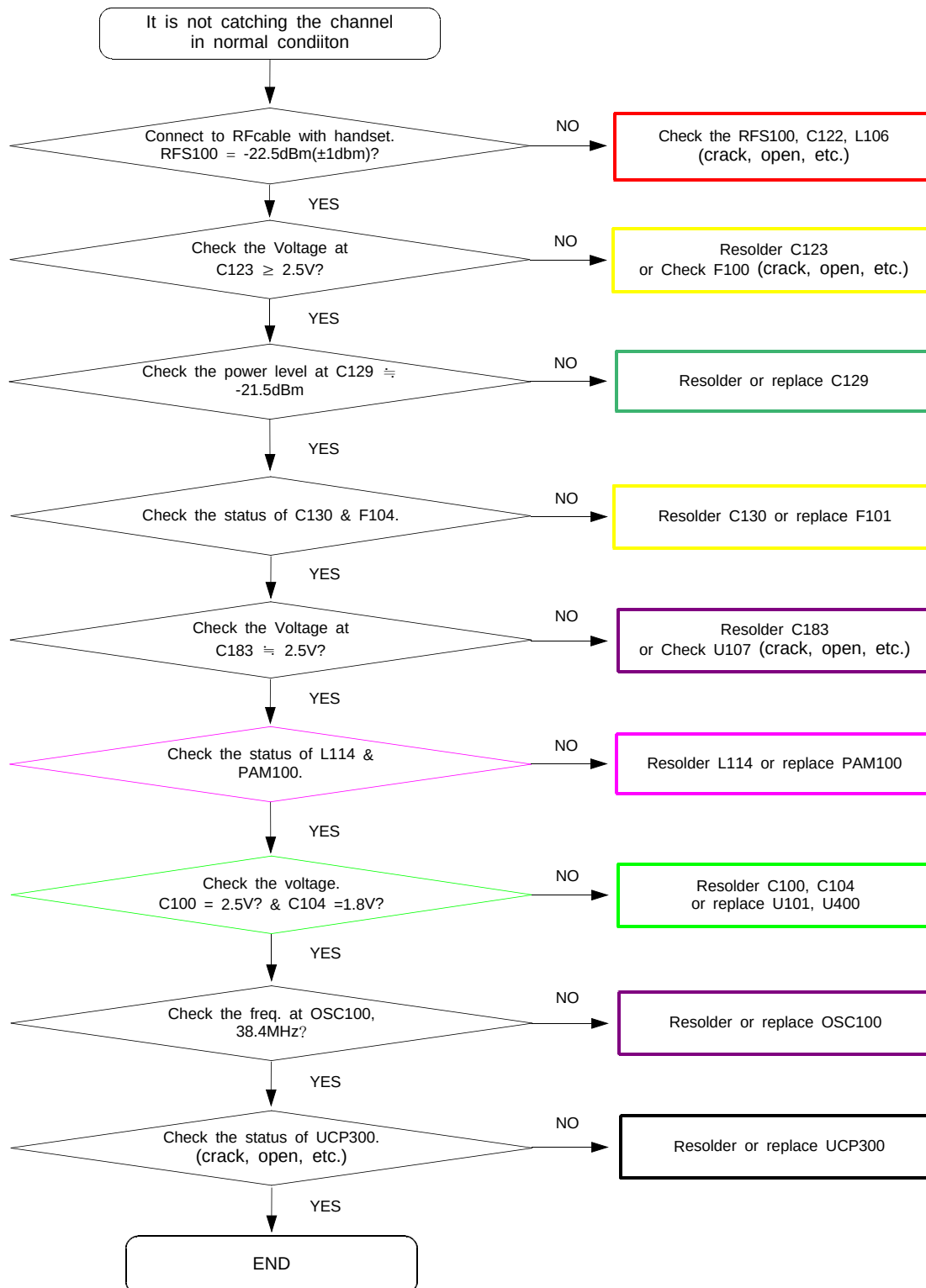


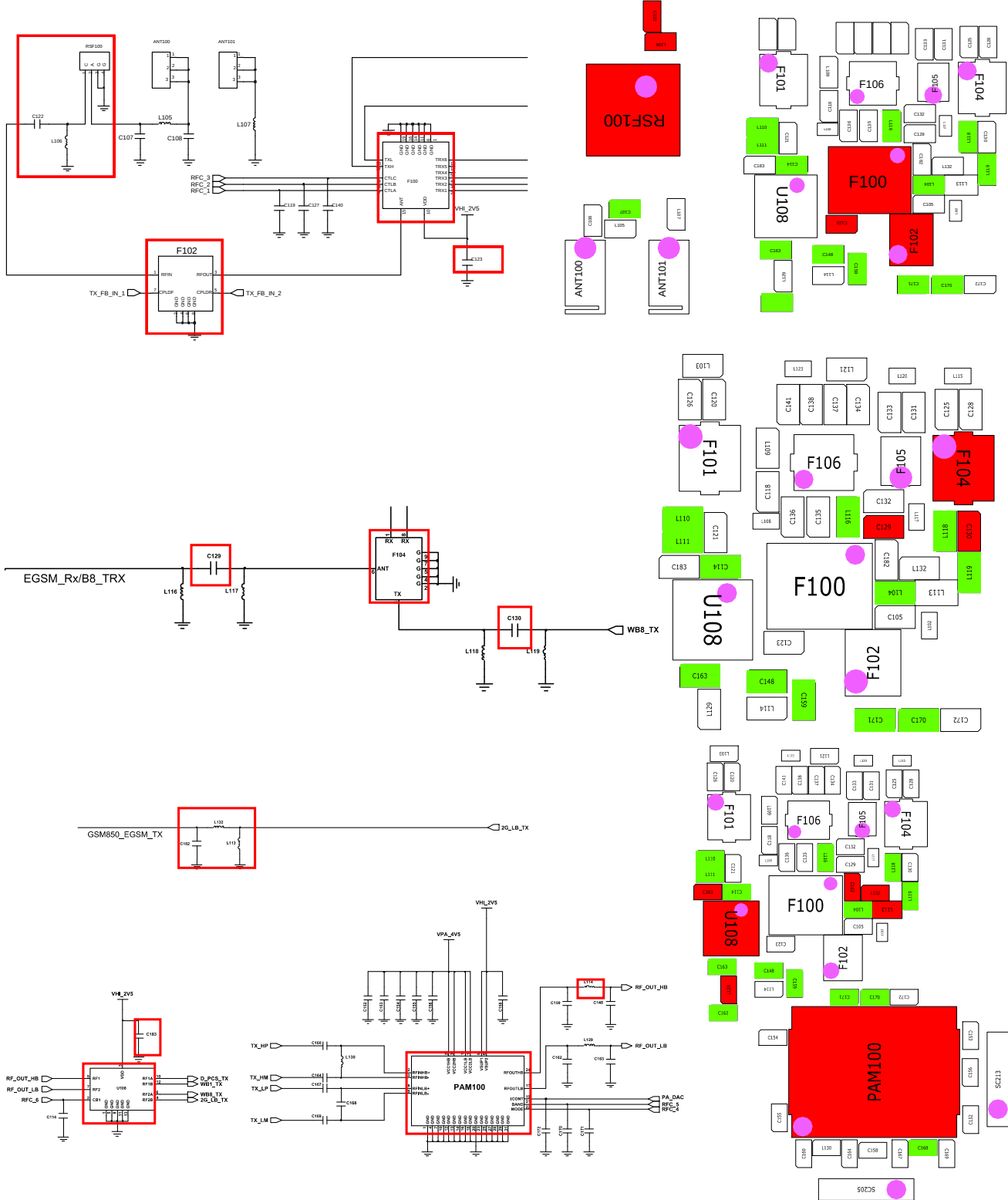
8-3-21. WCDMA BAND1 Tx



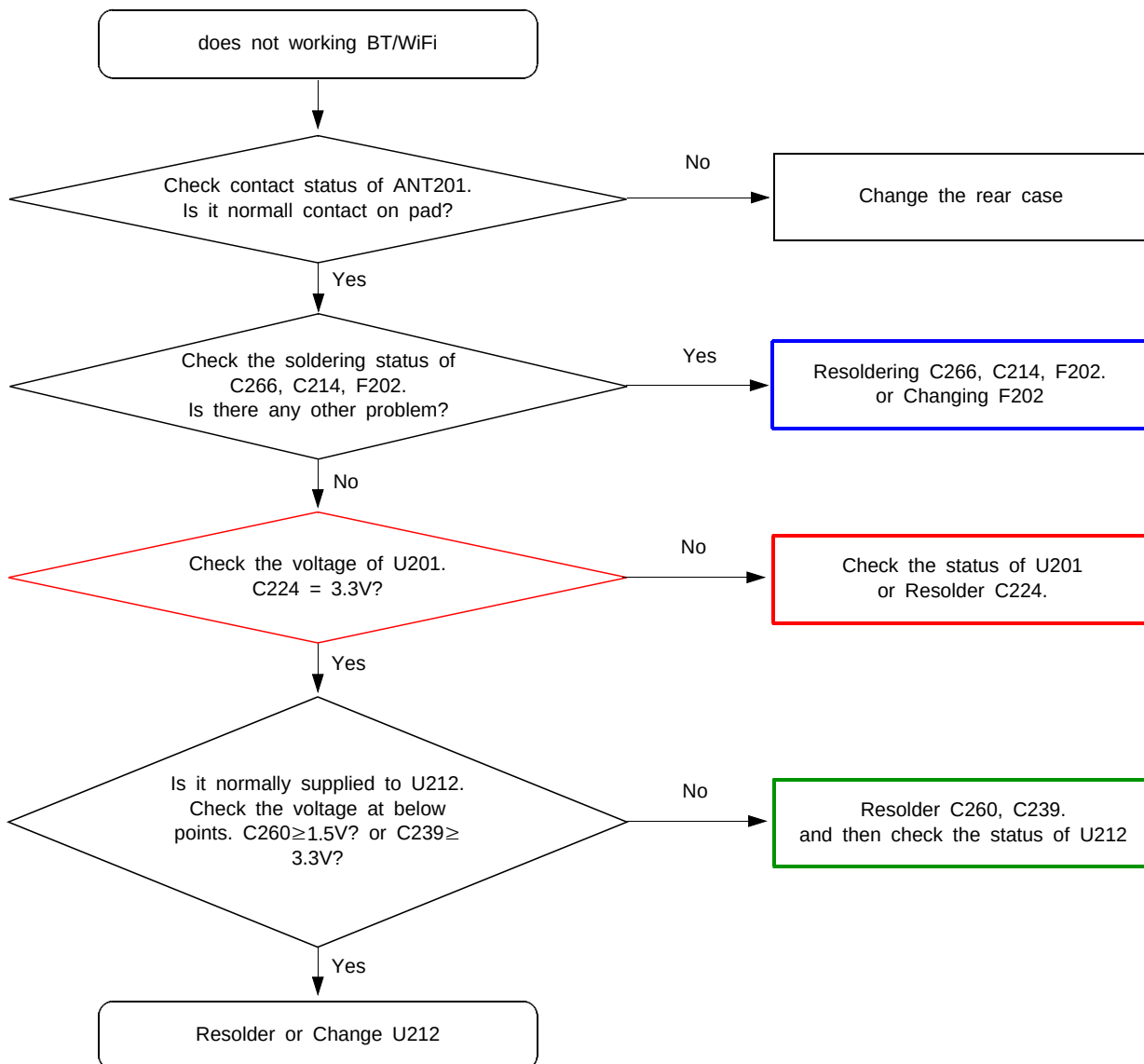


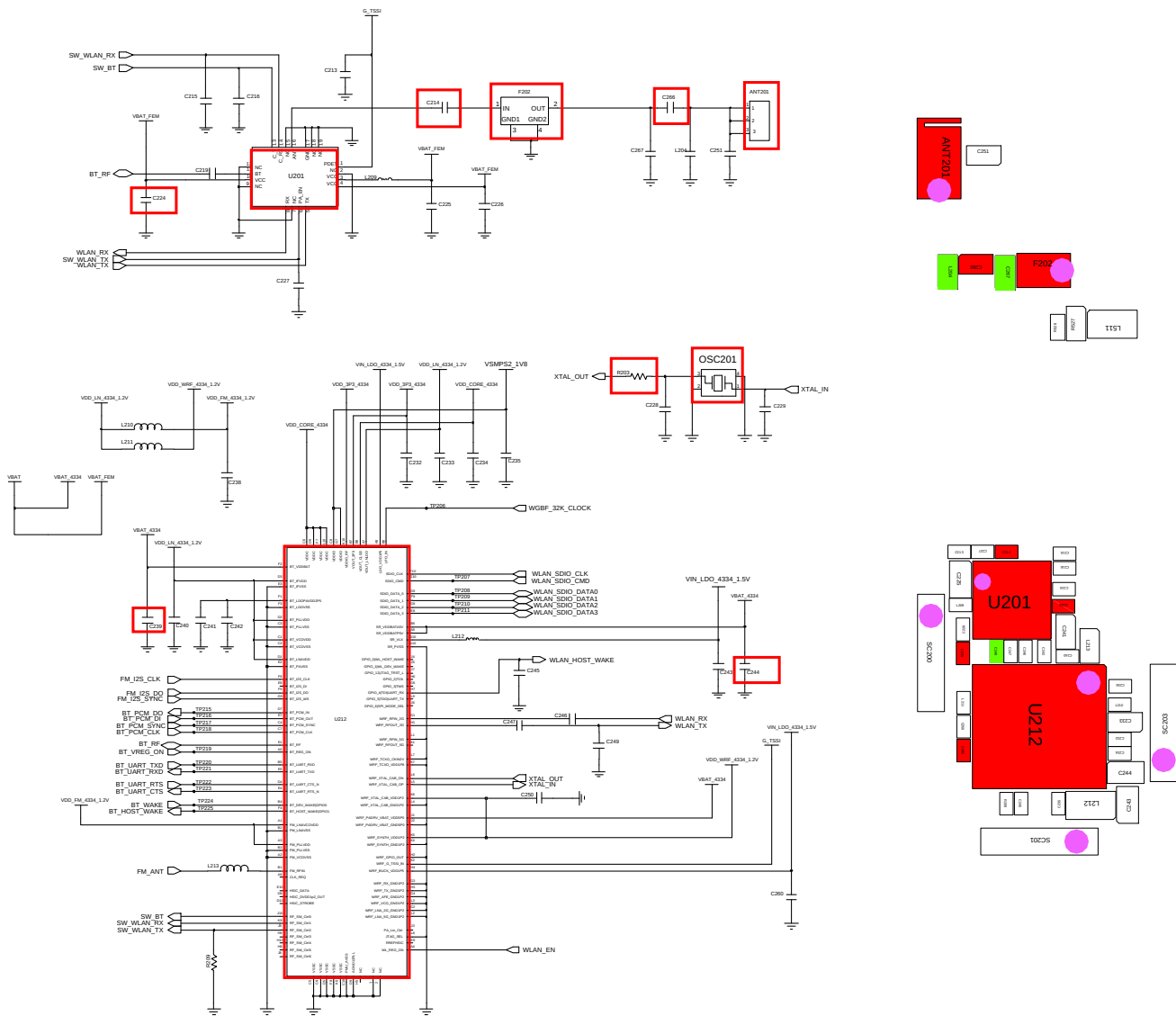
8-3-21. WCDMA BAND8 Tx



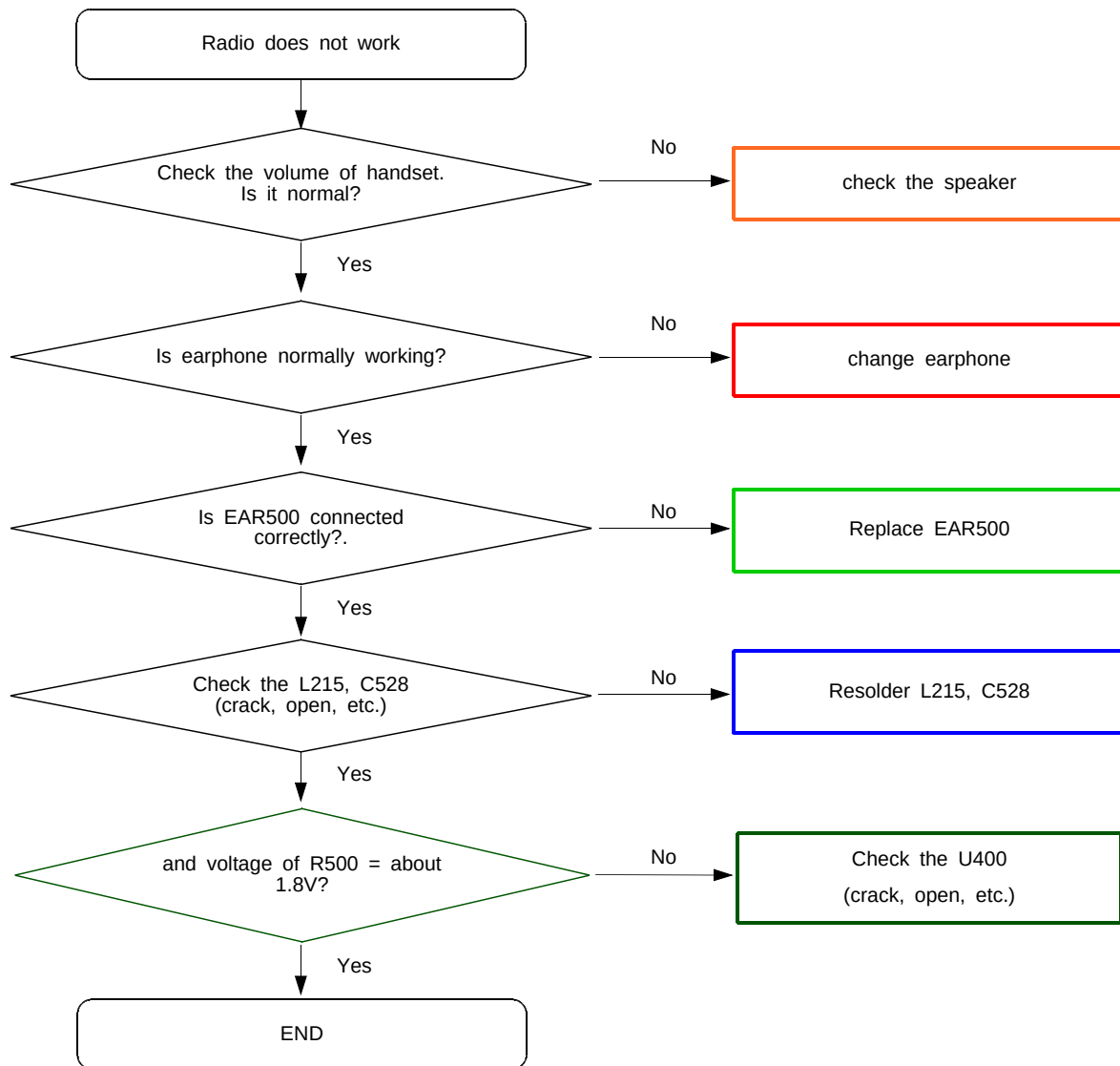


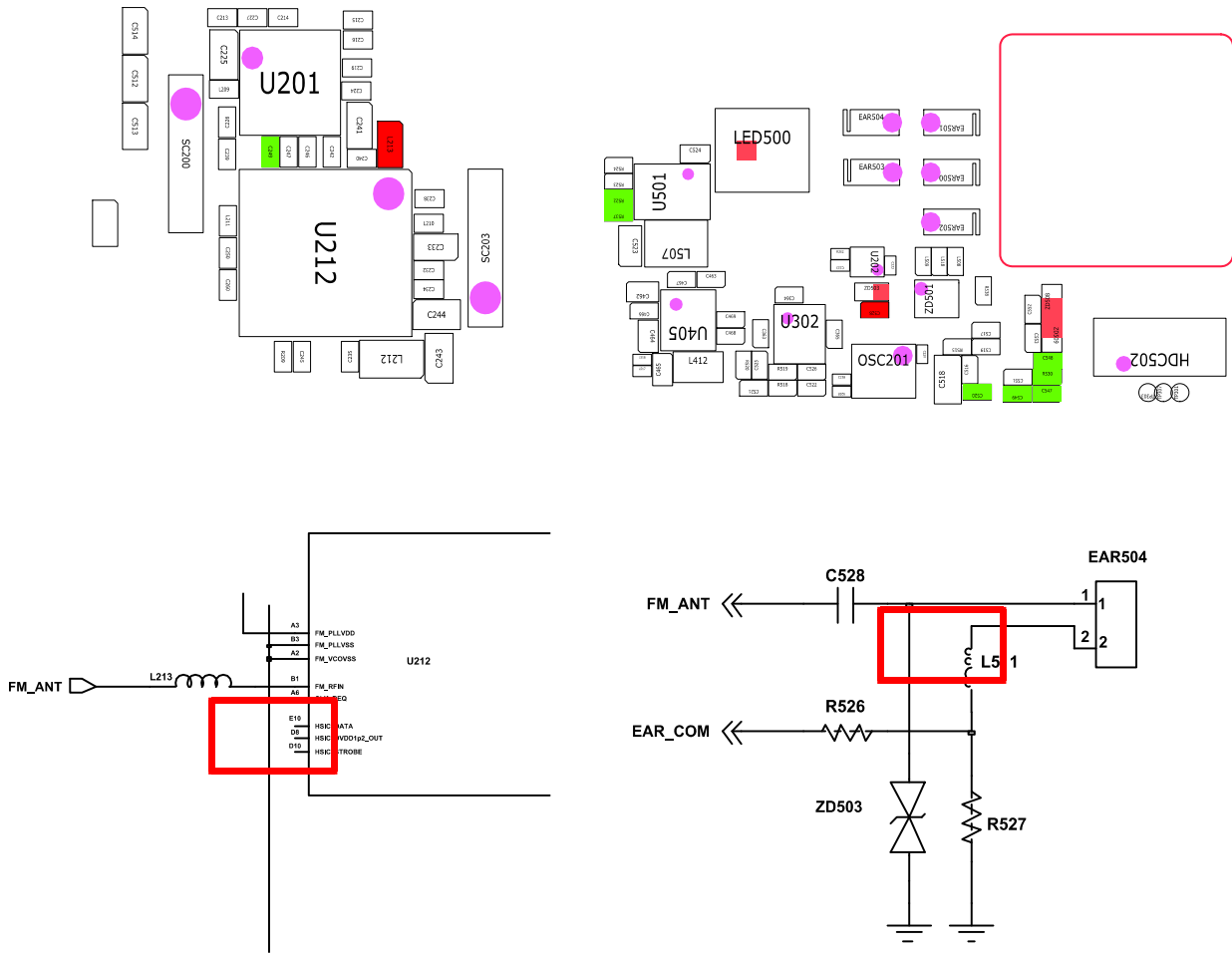
8-3-9. BT/WIFI





8-3-10. FM RADIO





8-6. Service Schematics

- NC Point(Top View)

● : NC

U301

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
A	●	●	○	○	○	○	○	●	●	●	●	●	●	●
B	●	○	○	○	○	○	●	●	●	●	●	●	●	●
C	●	○	●	○	○	○	●	●	●	●	●	●	●	●
D	●	●	●	●								●	●	●
E	●	●	●		○	○	○	○	○	○		●	●	●
F	●	●	●		○					○		●	●	●
G	●	●	○		○					○		●	●	●
H	●	●	●		○					○		●	●	●
J	●	●	●		○					○		●	●	●
K	●	●	●		○	○	○	○	○	○		●	●	●
L	●	●	●									●	●	●
M	●	●	●	○	○	○	●	●	●	●	●	●	●	●
N	●	○	●	○	○	●	●	●	●	●	●	●	●	●
P	●	●	○	○	○	○	○	●	●	○	●	●	●	●